


# A Hierarchical ReAct Multi-Agent Pipeline for Residential Appliance Load Scheduling Under Dynamic Electricity Pricing

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# Abstract

Households on dynamic electricity contracts can lower their bills by running flexible appliances in low-price hours, but coordinating several appliances under a volatile price curve and a household peak load limit is difficult without automation. This thesis follows Design Science Research to design and evaluate a hierarchical ReAct multi-agent system, built in LangGraph, that gives a household a daily appliance recommendation. An Orchestrator, an Appliance Profiler, a Price-Aware Scheduler and a Conflict Resolver work from the day's appliance plan, the base load, the peak limit and the prices, and a verification step returns flawed schedules for revision. The agents are given similar past days whose optimal schedules are known as worked examples, and an optimiser tool that they call, check and explain. The study asks how much the recommendations save, how often they are valid and what they cost to produce compared with a language model answering without tools (SQ1), and under which price patterns, household loads and peak limits the recommendations save more than, or as much as, existing home energy management tools available to households: cheapest-hours automations and the open-source optimiser EMHASS (SQ2).

The evaluation combines appliance cycles from the REFIT dataset with Finnish Nord Pool prices in a factorial benchmark of four price patterns, two household loads and two peak limits. The design was developed on 96 exploratory scenarios, and seven pre-registered hypotheses were tested on 96 held-out scenarios with fifteen-minute prices. The primary outcome is the saving against starting every appliance when it is ready. On the held-out scenarios, the pipeline with GPT-4o-mini produced a valid schedule in all 288 runs and saved 27.4 % of the cost, the same as the optimal schedule. It saved 15.4 percentage points more than the same model without tools, at an API cost of $0.0025 per recommendation. It saved 8.6 to 10.0 percentage points more than the cheapest-hours automations, whose schedules plan each appliance on its own and exceed the household peak limit on most days that need coordination, and the same as EMHASS within 0.1 percentage points. Five of the seven hypotheses were supported. The pipeline was not valid significantly more often than the model without tools, which was itself valid in 96.5 % of its runs, and its mean response time of 63.5 seconds missed the objective of 60 seconds after the language model service slowed down during the run; a later design iteration with fewer model calls halved the response time on the exploratory benchmark. The pipeline thus matches an optimizer in cost and adds an explanation of every schedule, while a simple peak-aware rule written for this study came within 0.1 percentage points of it without a language model.

*Keywords*

multi-agent system, ReAct, LangGraph, retrieval-augmented generation, home energy management, appliance scheduling, dynamic electricity pricing, Design Science Research

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# Foreword

Work in progress.

# Abbreviations

| | |
|---|---|
| **API** | Application Programming Interface |
| **DSR** | Design Science Research |
| **HEMS** | Home Energy Management System |
| **LLM** | Large Language Model |
| **MAS** | Multi-Agent System |
| **MCP** | Model Context Protocol |
| **MILP** | Mixed Integer Linear Programming |
| **Nord Pool FI** | Nord Pool Finland Bidding Zone |
| **RAG** | Retrieval-Augmented Generation |
| **ReAct** | Reasoning and Acting |
| **REFIT** | Personalised Retrofit Decision Support Tools for UK Homes using Smart Home Technology (the research project that produced the dataset) |
| **SLM** | Small Language Model |
| **SLR** | Systematic Literature Review |

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# 1. Introduction

Smart meters, dynamic electricity contracts and a growing share of renewable generation have changed what a household pays for electricity from hour to hour. A household on a spot-price contract can lower its bill by running flexible appliances, such as the washing machine or the dishwasher, in the day's low-price hours. Without automation this is hard when several appliance cycles have to fit a price curve that changes every day and the household must also stay under a limit on its total power.

Finland is a clear case. Since October 2025, Finnish consumers on spot-price contracts pay prices set for every 15 minutes on the Nordic spot market (Nord Pool FI bidding zone), and prices within a day vary widely. From 2023 to 2026, the most expensive hour of a day cost a median of 3.1 times the cheapest on days without near-zero prices, and one day in five had at least one hour with a negative price (computed from Nord Pool FI prices; Pakastin Oy, 2026). Such extreme prices measurably change household consumption and affect the distribution network (Haakana et al., 2025). Language models have recently been applied to the same problem: in the first Home Energy Management Systems (HEMS) built on large language models, agents schedule household appliances, a task that earlier needed hand-written rules or an optimization solver (El Makroum et al., 2025; Jonah et al., 2026). These first systems have documented limitations, which this thesis takes up.

## 1.1 Background and motivation

HEMS have mostly been built on mathematical optimization, rule-based schedulers, or machine learning models trained on past consumption (Beaudin & Zareipour, 2015). These approaches work well for a precisely specified problem, but they take expert effort to configure, and a rule or a trained model does not carry over to price conditions it was not designed for.

Large Language Models (LLMs) can be used as general reasoning engines, which gives HEMS design another option. Yao et al. (2023) introduce ReAct, a Thought-Action-Observation loop in which the model alternates between reasoning and tool calls, and show that it solves multi-step tasks that need information looked up along the way. For household scheduling, a ReAct agent can read the price forecast, decide which appliances can be shifted, find the cheapest time window for each and check the result against the household's constraints, and every step of that reasoning is recorded.

A secondary motivation concerns sustainable AI deployment. The energy needed per inference differs by orders of magnitude between models and tasks (Luccioni et al., 2024), and serving a small language model (SLM) of about 7 billion parameters has been estimated to be 10 to 30 times cheaper in latency, energy and computation than serving a model of 70 to 175 billion parameters (Belcak et al., 2025). Whether such smaller models

are good enough for a given agentic task is an empirical question. Each recommendation requires several calls to a language model, which costs money and energy. The inference cost of each design, measured in tokens and response time, is therefore reported alongside the saving it achieves. A design that reaches the same saving with fewer model calls is cheaper to run and easier to move to a small local model later.

## 1.2 Research problem

In the hierarchical multi-agent system of El Makroum et al. (2025), only one of three evaluated models coordinated all appliances, and the evaluation covers a single day. Reviewing that work, Jonah et al. (2026) list multi-appliance coordination failures, scheduling on price alone, a single-day evaluation horizon and untested constraint safety as open problems. Their own single-agent system is evaluated for one household over a few weeks, and they recommend that every schedule be validated independently before it is executed. Neither study varies the price conditions. It is also unclear what a language model agent adds over the tools households can already use, such as cheapest-hours automations and open-source optimizers, and whether an agent can match an optimizer when several appliances have to be staggered. Two techniques used in other LLM systems could help but have not been tried for household scheduling: giving the model retrieved examples of solved problems (retrieval-augmented generation, RAG), and giving it a solver as a tool.

The artifact studied in this thesis gives each household a daily recommendation. Every day it receives the household's appliance plan for that day, namely which appliance cycles are to run and when each is ready and must be finished, together with the household's base load, its peak load limit and the day's electricity prices. It returns a start time for each appliance and a short explanation. To address the limitations above, the study is guided by one main research question and two sub-questions:

> **RQ:** How can a hierarchical ReAct multi-agent system be designed to give households reliable, cost-saving daily appliance recommendations under dynamic electricity prices?
>
> **SQ1 (value and cost):** How much do the pipeline's recommendations save, how often are they valid, and what do they cost to produce, compared with a language model that answers without tools?
>
> **SQ2 (conditions):** Under which price patterns, household loads and peak limits do the recommendations save more than, or as much as, existing home energy management tools available to households, namely cheapest-hours automations and the EMHASS optimizer?

The main RQ states the design aim: recommendations that are reliable, meaning that they respect every appliance window and the household peak limit, and that save money against starting every appliance when it is ready. The pipeline studied is the final design:

four ReAct agents that are shown how similar past days were solved best and that hand the search to an optimiser tool, then check the answer and explain it. SQ1 concerns what its recommendations are worth and what they cost. It tests the premise of the ReAct approach, that reasoning grounded in tool observations does better than unaided generation, by comparing the pipeline with the same language model answering directly, and it checks the response time and API cost of each recommendation against the design objectives. SQ2 concerns where the system is worth using. Its ”existing home energy management tools” are real, freely available software that a household can install without programming: the cheapest-hours automations shared for Home Assistant and Shelly smart relays, which switch a device on in the day’s cheapest hours, and the open-source optimizer EMHASS, which computes a schedule for the whole household. SQ2 compares the recommendations with these tools and asks how the comparison depends on the day’s prices, on how many appliances the household runs, and on how tight its peak limit is.

## 1.3 Design Science Research approach

This study follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). The study builds a software artifact, the hierarchical ReAct multi-agent system, and evaluates it over several iterations, which is the kind of work DSR is meant for. As Hevner et al. (2004) explain, DSR produces knowledge about a design problem and its solution by building and evaluating an artifact. The artifact is the main research contribution. A Systematic Literature Review (SLR) forms DSR Phase 1: it confirms the research gap and grounds the architectural decisions in prior work.

## 1.4 Structure of the thesis

This thesis is organized into seven chapters. Chapter 2 presents the literature review covering HEMS concepts, LLM-based agents for energy management, the ReAct framework, multi-agent LLM systems, retrieval and solver tools for agents, current agentic AI trends, and the Finnish energy market context. Chapter 3 describes the Design Science Research methodology and evaluation strategy. Chapter 4 presents the system design and implementation. Chapter 5 presents the results on an exploratory and a held-out benchmark of 96 scenarios each. Chapter 6 discusses findings, limitations, and future work. Chapter 7 concludes the thesis. Appendix A repeats the comparison with UK prices of the period in which the appliance data were recorded, and Appendix B reports the evaluation of an earlier design that compared the pipeline with a single agent.

# 2. Literature Review

This chapter reviews the background and prior empirical work on which the design and evaluation of the multi-agent system rest. It covers eight topics and ends with the research gap that the research questions address.

## 2.1 Home energy management systems

A Home Energy Management System (HEMS) is a software or hardware system that monitors, controls, and optimizes residential electricity consumption to reduce cost, support grid stability, and improve energy efficiency (Beaudin & Zareipour, 2015). The core scheduling problem is the assignment of flexible appliance loads to time windows where electricity prices are lowest, while ensuring that total household load at any moment does not exceed safe operational limits. A load is flexible when its start time can be shifted without violating user preferences or physical constraints.

HEMS scheduling approaches fall into three broad groups. Rule-based systems apply expert-defined heuristics, for example starting an appliance when the price falls below a threshold or in the cheapest hours of a window. They are simple to run and easy to follow, because each decision follows from one stated rule, but each rule covers only the situation it was written for. A rule that picks the cheapest hours for one device does not see the other devices, so it cannot keep the household under a limit on its total power, and a new requirement means writing and testing a new rule.

Mathematical optimization methods such as Mixed Integer Linear Programming (MILP) compute the best schedule for a precisely specified problem. Sou et al. (2011) formulate appliance scheduling with binary start variables, uninterrupted cycles and time windows, and Moradzadeh and Tomsovic (2013) formulate residential energy management as a two-stage optimization that respects the operational constraints of the distribution network. Such methods are optimal for the model they solve, but every constraint must first be expressed in mathematical form, so a household's preferences have to be translated into variables and constraints by someone who can do so, and a new kind of constraint needs a new formulation. Their output is a schedule, not a reason for it.

Learning-based methods acquire a scheduling policy from data, as in the deep reinforcement learning approach of Mocanu et al. (2019), which is trained on measured building data and then optimizes schedules on-line. They adapt to the household they are trained on, but a change in the task, such as a new appliance or a new tariff, generally requires new training, and the learned policy is hard to inspect.

Optimization and learning-based controllers have been described as black boxes that are hard for users to understand and trust, which has led to work on explainable HEMS,

for example by distilling a reinforcement-learning policy into a decision tree that can be explained to end-users (Gokhale et al., 2024). Zharova et al. (2024) argue that users need to understand why and how a recommendation system reaches its recommendations before they trust and accept them, and add a dedicated explainability agent to a multi-agent system that recommends when to run household appliances. Explanation in terms a non-expert understands, and adaptation to new constraint types without a new rule, formulation or training run, are therefore the two properties that a language model agent could add to these approaches.

Rule-based and optimization approaches are both available to households today as open-source tools. Among Home Assistant users in Finland and the other Nordic countries, a widely shared approach is to automate devices with community blueprints that switch a device on during the cheapest hours of a user-defined time window, based on Nord Pool day-ahead prices (Kert, n.d.). Related tools choose the block of consecutive hours with the lowest average price before a set end time, such as the Nordpool Planner integration for Home Assistant (dala318, n.d.), or switch Shelly smart relays on in the cheapest hours of a period, as the shelly-porssisahko script widely used in Finland does (jisotalo, n.d.). Such rules handle each device separately and do not consider the household's total load. EMHASS (Energy Management for Home Assistant) is an open-source optimizer that schedules deferrable loads, batteries and photovoltaic production against price forecasts with linear and mixed-integer programming (EMHASS contributors, 2026). It supports a time window for each load, a limit on the power drawn from the grid, and loads defined by a fixed power profile. Both kinds of tool require the household to describe each appliance in structured form, namely its power, run time and time window. OpenHEMS, a Home Assistant add-on, schedules heavy appliances under a maximum power but targets fixed off-peak tariffs rather than spot prices (OpenHEMS contributors, 2026). This thesis uses the cheapest-hours rules and EMHASS as reference points for the agents (Section 3.3).

## 2.2 LLM-based agents for energy management

Studies applying LLMs to energy management have appeared since 2023, and two recent ones are close to this thesis. El Makroum et al. (2025) present a hierarchical agentic architecture for home energy management in which an orchestrator agent delegates to three specialist agents that follow the ReAct pattern. The household has three flexible loads: a washing machine (2.0 kW for 120 minutes), a dishwasher (1.8 kW for 90 minutes) and an electric vehicle charger (7.4 kW for 360 minutes). The agents receive Austrian day-ahead prices from ENTSO-E for one day, 14 October 2025, and are run with three open-weight models, Llama 3.3 70B, Qwen 3 32B and GPT-OSS 120B. The evaluation consists of 75 runs: 15 single-appliance runs, 15 multi-appliance runs and 45 runs in which the user asks analytical questions about the schedule. Each schedule is compared with the optimum of a MILP model rather than with an unscheduled baseline, so the study reports how often the agents reach the optimal cost, not how much they save. All three models scheduled single appliances optimally, but only Llama 3.3 70B coordinated all appliances in every

multi-appliance scenario, and the analytical questions were answered unreliably unless the agents were given explicit guidance. The authors describe the study as a proof of concept: one household, one day and one price series, without thermal loads such as heat pumps, electric heating and hot water storage. They name pilot deployments and evaluation across electricity markets and household types as future work. Of the published systems, this design is the closest to the pipeline of this thesis.

Jonah et al. (2026) take the opposite design choice: a single tool-calling ReAct agent that turns a household's request into appliance schedules, using half-hourly dynamic prices from the United Kingdom (Octopus Agile), weather and photovoltaic forecasts and household demand data. They test five language models, among them Claude Sonnet 4.6, GPT-4o-mini and Qwen 3. Their agents capture 96.7 to 98.0 % of the savings available between an off-peak timer and a MILP optimum and do better than rule-based policies. Their failure analysis is the more important result for this thesis. The models sometimes fabricated schedules or failed to commit one, and in some cases a model explained a deadline correctly and still committed a schedule that violated it. Actions parsed from text were less reliable than native function calls. Building on El Makroum et al., they list multi-appliance coordination failures, scheduling on price alone, a single-day evaluation horizon and untested constraint safety as open problems, and they recommend that an independent deterministic validator check every schedule against duration, deadline and power constraints before it is executed. Their evaluation covers one household over twelve tariff days and a one-week deployment. Section 2.8 sets out how these open problems shape the design and evaluation of this thesis.

## 2.3 The ReAct framework

In the ReAct (Reasoning and Acting) framework of Yao et al. (2023), an LLM agent interleaves reasoning with actions instead of first planning everything and then acting. Each step has three parts: a Thought, which reasons about the current state and the next action; an Action, which is a structured call to an external tool or API; and an Observation, which is the result the tool returns. The loop repeats until the agent reaches a final answer.

Yao et al. (2023) evaluate ReAct on four benchmarks of two kinds. Two are knowledge-intensive reasoning tasks, in which the agent can query a Wikipedia API: HotpotQA, multi-hop question answering, and FEVER, fact verification. Two are interactive decision-making tasks: ALFWorld, a text-based game in which the agent carries out household chores in a simulated home, and WebShop, in which it shops on a simulated website to match a customer's instructions. Their baselines are built by removing parts of the ReAct trajectories: standard prompting without reasoning or actions, chain-of-thought prompting with reasoning but no actions, and acting alone without reasoning, so that each comparison isolates the contribution of reasoning or of acting. With PaLM-540B, ReAct reached 27.4 exact match on HotpotQA, slightly below chain-of-thought (29.4), and 60.9 % accuracy on FEVER, above it (56.3 %), while the best results on both came from combining the two. On the interactive tasks the advantage was clear: on ALFWorld the best ReAct trial reached a 71 % success rate against 45 %

for acting alone, and on WebShop 40.0 % against 30.1 %. The gains were largest where the agent has to act on information it looks up during a multi-step task. Household scheduling is such a task: the agent looks up prices, loads and costs while it builds the schedule. The trace also makes the result explainable, because it records the reasoning behind every tool call, so it shows why an appliance was given a particular start time.

## 2.4 Multi-agent LLM systems

Multi-agent LLM systems have been proposed as a solution to the context window and coordination limitations of single-agent designs. Han et al. (2024) discuss the challenges and open problems of LLM multi-agent systems, among them task allocation, robust reasoning through iterative debate, the management of complex and layered context information, and memory management. Role specialization, in which each agent is responsible for one well-defined subtask and has access only to the tools that subtask needs, is one response to these challenges and the design principle followed in this thesis.

Hong et al. (2024) demonstrate role-based multi-agent orchestration for software development with MetaGPT, in which agents act as product manager, architect, project manager, engineer and quality-assurance engineer. Encoding standard operating procedures and requiring structured intermediate outputs, exchanged through a shared message pool, reduces the inconsistencies that arise when language model calls are simply chained, and MetaGPT reported state-of-the-art results on code-generation benchmarks at the time. Structured hand-offs between specialists address the same kind of problem as the multi-appliance coordination failures reported for LLM-based HEMS (El Makroum et al., 2025).

The move from one agent to a hierarchy of agents is motivated by three problems of a single agent that handles a whole task. First, its context grows with every tool call and observation, and the model has to keep track of all of them, while a specialist sees only the information of its own step. Second, a single agent has access to every tool, so it can call any of them at any point, while a specialist is limited to the few tools its subtask needs, which narrows the choices it can get wrong. Third, the intermediate results of a single agent stay inside its own reasoning, while a hierarchy passes them between agents as structured outputs, which can be checked before the next step (Hong et al., 2024). In a hierarchical design an orchestrator splits the task, hands each part to a specialist and combines the results; the survey of Guo et al. (2024) describes such layered structures alongside centralized and decentralized ones as ways in which the agents of an LLM-based multi-agent system communicate. The hierarchy has a price: more model calls per task, hand-offs that can lose information, and the need to decide in advance how the task is split. For household scheduling the split follows the structure of the problem: which appliances can move, when each is cheapest, and how they fit under the peak limit together. The verification step between the specialists is where the independent check recommended by Jonah et al. (2026) fits.

## 2.5 Retrieval and solver tools for LLM agents

Two techniques extend what a language model agent can do without retraining it. Retrieval-augmented generation (RAG) gives the model relevant documents from an external store at the time of the request, so that its answer draws on knowledge it was not trained on and the store can be updated independently of the model (Lewis et al., 2020). For a planning task, the retrieved material can be solved examples of similar problems, which show the model what a good solution looks like. Whether such examples improve the search of a multi-step plan, rather than only its form, is an empirical question.

The second technique keeps the search out of the language model altogether. In OptiGuide, a language model answers questions about a supply chain by writing and running code against an optimization solver and explaining the result; the authors stress that the aim is to use solvers in tandem with language models rather than to replace them (Li et al., 2023). A household scheduling agent can be built the same way: the agents collect the household's constraints, call a solver, check its answer and explain it, while the solver guarantees the best schedule. Neither technique has been evaluated for household appliance scheduling. This thesis studies both as extensions of the pipeline (Section 4.4).

## 2.6 Agentic AI trends in 2025–2026

Three recent developments in agentic AI bear on this thesis. The first is the move from single agents to systems in which several LLM-based agents with distinct roles cooperate, a research area that has grown quickly since 2023 (Guo et al., 2024; Wang et al., 2024). Two surveys describe how the profiles, memory, planning and communication of such agents are organized, and they identify coordination between agents and the evaluation of multi-agent systems as open challenges. The idea is that a complex task can be divided among specialists, each with a smaller context and fewer tools than a single agent that handles the whole task.

The second is the Model Context Protocol (MCP), introduced by Anthropic in November 2024 and donated to the Linux Foundation's Agentic AI Foundation in December 2025, which has become a common standard for connecting agents to tools (Anthropic, 2024). MCP gives agents one model-independent interface to external tools and data sources, so each tool no longer needs its own integration code. The LangChain MCP adapters library makes MCP tools usable by LangGraph agents, so the price API and data reader of this thesis could be offered as reusable MCP servers.

The third is a growing set of design patterns for such agents. Liu et al. (2025) catalogue eighteen architectural patterns for agents built on foundation models, three of which match designs in this thesis: role-based cooperation, in which agents with assigned roles divide the work, describes the multi-agent pipeline; one-shot model querying, in which the model produces its answer in a single step, describes the LLM-without-tools baseline; and human reflection, in which humans give feedback on the agent's plan, describes a human-in-the-loop (HITL) checkpoint. Guidelines for human-AI interaction

likewise recommend that a system make clear why it did what it did and support efficient correction by the user (Amershi et al., 2019). The artifact therefore has an optional HITL checkpoint, a LangGraph interrupt node at which the Orchestrator shows the proposed schedule to the user for approval before it is finalized. The checkpoint was not enabled in the reported evaluations, which run without human intervention.

## 2.7 Finnish energy market context

Finland belongs to the Nordic spot electricity market (Nord Pool) and has priced electricity in 15-minute intervals since October 2025, which makes its residential market one of the most finely priced in Europe. At the end of 2024, about 33 % of all Finnish electricity contracts were dynamic-price contracts, up from 9 % in 2021; these pass hourly and quarter-hourly price changes straight to the consumer (Energy Authority, 2025).

Prices within a day vary a great deal. In the Nord Pool FI day-ahead prices from 2023 to 2026, the difference between the cheapest and the most expensive hour of a day had a median of 5.7 c/kWh excluding VAT and exceeded 10 c/kWh on 28 % of days. On days without near-zero prices the most expensive hour cost a median of 3.1 times the cheapest, and on 19 % of days at least one hour had a negative price (computed from Pakastin Oy, 2026). Extremes are far larger: the highest hourly price in the period, 189.6 c/kWh on 5 January 2024 at 19:00, was about one hundred times the median night-time price of 1.8 c/kWh. On 24 November 2023 the price fell to $-500$ EUR/MWh for half a day, and households with spot-price contracts noticeably increased their consumption (Haakana et al., 2025). At the same time, an estimated 7 to 15 % of Finnish households are energy poor (Aalto University, 2026). Shifting household demand in Finland therefore matters for the power system as well as for household bills.

## 2.8 Literature synthesis and research gap

Three findings come out of the reviewed literature. First, HEMS scheduling is a multi-step, multi-constraint optimization problem. Rules and optimizers solve it for the cases they were written or formulated for but do not explain their decisions, and language model agents have been proposed because they can take constraints stated in words and explain what they did (Section 2.1). Second, the first LLM-based HEMS implementations, both hierarchical and single-agent, have documented limitations in multi-appliance coordination, in scheduling on price alone, in the safety of committed schedules, and in evaluation scope (El Makroum et al., 2025; Jonah et al., 2026). Third, the existing home energy management tools that households can install range from cheapest-hours rules that plan each device alone to optimizers such as EMHASS, and retrieval of solved examples and solver tools are established ways to strengthen language model agents (Lewis et al., 2020; Li et al., 2023). No published HEMS study measures what an agent adds over these existing tools across several price patterns, or whether retrieval or a solver tool lets an agent reach the saving of an optimizer. Table 2.1 links

each limitation of the prior work to the design or evaluation choice of this thesis that responds to it, and to the research question that tests it.

**Table 2.1 Limitations of prior work and how this thesis responds to them.**

| Limitation in prior work | Response in this thesis | Tested in |
|---|---|---|
| Multi-appliance coordination fails for some models (El Makroum et al., 2025; Jonah et al., 2026) | A Conflict Resolver, retrieval of solved days, an optimiser tool, and a household peak limit that forces appliances to be staggered on heavy-load days | SQ1, SQ2 |
| Committed schedules can violate constraints the model reasoned about correctly; an independent check is recommended (Jonah et al., 2026) | A deterministic verification step checks every schedule for conflicts, missing appliances and window violations, and returns it for revision | SQ1 (validity) |
| Evaluation on a single day or a few days, one household, one price series (El Makroum et al., 2025; Jonah et al., 2026) | A factorial benchmark of four price patterns, two household loads and two peak limits, built from 18 REFIT households, and a held-out benchmark with pre-registered hypotheses | SQ2 |
| Scheduling on the price signal alone (Jonah et al., 2026) | Partly addressed: every schedule must also keep base load plus appliances under a peak limit, with a strict and a realistic limit compared; the objective remains the electricity cost, and other signals such as carbon intensity are not added | SQ2 |
| No comparison with existing home energy management tools | Comparison with cheapest-hours automations for Home Assistant and Shelly relays and with the optimizer EMHASS | SQ2 |
| Rules and optimizers do not explain their decisions (Gokhale et al., 2024; Zharova et al., 2024) | The Orchestrator explains every recommendation and states the result of the final peak-load check | design |

This thesis addresses that gap with three contributions. The first is the Multi-Agent Pipeline, a hierarchical ReAct multi-agent system in LangGraph that gives households daily appliance recommendations, using a library of solved days for retrieval and an optimiser tool; the design iterations that led to it are documented. The second is a measure of what its recommendations save, how often they are valid and what they cost to produce compared with a language model answering without tools (SQ1). The third is a map of the conditions, four price patterns, two household loads and two peak limits, under which the recommendations save more than, or as much as, existing home energy management tools, cheapest-hours automations and EMHASS (SQ2). The hypotheses were fixed in advance and tested on a held-out benchmark.

# 3. Research Methodology

This chapter describes the research methodology and explains how Design Science Research is applied in this thesis. It presents the rationale for the approach and the mapping of its phases to the thesis, the theories on which the design of the artifact draws, and the evaluation strategy: the systems compared, how the data were collected, and how they were analysed.

## 3.1 Design Science Research methodology

This thesis follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). DSR designs and evaluates artifacts that solve practical problems. As Hevner et al. (2004) argue, design science creates and evaluates IT artifacts for identified organizational problems and produces knowledge by building and testing them. The main activity of this thesis is building and evaluating the multi-agent system, so DSR fits it.

Artifacts in DSR are classified into four types: constructs, models, methods, and instantiations (March & Smith, 1995). In this thesis, the artifact is an instantiation, that is, a working software implementation of a hierarchical multi-agent scheduling pipeline. It is supported by architectural models (the LangGraph state graph design) and constructs (the agent role taxonomy: Profiler, Scheduler, Resolver, Orchestrator). Table 3.1 maps the six DSR phases defined by Peffers et al. (2007) to the activities and outputs of this thesis.

**Table 3.1 DSR phase mapping.**

| Phase | DSR step | Activity in this thesis | Output |
|---|---|---|---|
| 1 | Problem identification | SLR confirming no multi-agent ReAct HEMS addresses multi-appliance coordination across multiple price patterns | Gap statement; confirmed RQs |
| 2 | Define objectives | Specify measurable success criteria: saving against no scheduling, validity, conflict resolution rate, latency, token use | Objective specification |
| 3 | Design and development | Implement Appliance Profiler, Price-Aware Scheduler, Conflict Resolver and Orchestrator in LangGraph; add a case library for retrieval and an optimiser tool | Working multi-agent artifact |
| 4 | Demonstration | Run pipeline on five representative scenarios; confirm valid cost-reducing schedules | Demonstration results |
| 5 | Evaluation | Exploratory run on 48 hourly-price scenarios; two pre-registered confirmatory runs on held-out fifteen-minute scenarios (Q, Q2); comparison with existing home energy management tools; Wilcoxon and TOST tests | Quantitative evaluation results |
| 6 | Communication | Write findings, design guidelines, limitations, and future work | Complete thesis |

## 3.2 Theoretical foundations

Design science research grounds the design of an artifact in kernel theories, the existing theories from which its design principles are derived, and positions its contribution against them (Gregor & Hevner, 2013). Four bodies of work serve this role here. The first is demand flexibility under time-varying prices: a household lowers its cost by moving flexible loads to low-price periods, within the time windows its members accept and the power limits of its connection. This is the problem that rule-based and optimization-based HEMS address (Section 2.1), and the MILP formulation of Sou et al. (2011) states it precisely. It defines what a good recommendation is: the cheapest schedule that respects every window and the peak limit.

The second is reasoning grounded in tool observations. ReAct (Yao et al., 2023) holds that a language model solves multi-step tasks more reliably when it alternates between reasoning and actions whose results it observes, instead of producing an answer from its parameters alone. It yields the design principle that every number an agent uses comes from a deterministic tool, and the LLM without tools is the baseline that tests it (H1a). The third is task decomposition with role specialization in multi-agent systems (Hong et al., 2024; Guo et al., 2024): a task split into specialist roles with structured hand-offs keeps each agent's context and tool set small and makes intermediate results checkable (Section 2.4). It yields the four roles of the pipeline and the verification step between

them.

The fourth is the reuse of solved problems. Case-based reasoning solves a new problem by retrieving similar past cases and adapting their solutions (Aamodt & Plaza, 1994), and retrieval-augmented generation supplies a language model with such material at the time of the request (Lewis et al., 2020). Together with the use of a solver as a tool (Li et al., 2023), it yields the case library and the optimiser tool of the final design. Guidelines for human-AI interaction (Amershi et al., 2019) add the requirement that the system explain what it did and state when a constraint is not met, which the Orchestrator's summary implements.

The evaluation follows the Framework for Evaluation in Design Science Research (FEDS) of Venable et al. (2016). All evaluations are artificial, on simulated household days, and they move from formative to summative. The design iterations were formative evaluations on an exploratory benchmark, and the pre-registered run on a held-out benchmark is the summative evaluation. Of the strategies FEDS describes, this corresponds to Technical Risk and Efficacy, which is suited to an artifact whose main risk is whether it works technically before it is evaluated with users. A naturalistic evaluation with real households is left to future work (Section 6.5).

## 3.3 Evaluation strategy

The evaluation is a controlled simulation that needs no sensors or hardware. Its inputs come from public datasets and APIs, and every scenario is built from measured appliance data and historical prices. The comparison is paired: every system receives the same scenarios and is scored by the same function.

**Systems compared.** The artifact is the *Multi-Agent Pipeline* with GPT-4o-mini (Chapter 4). Its Scheduler and Resolver receive the two most similar solved days from a case library as worked examples, and the Scheduler has a tool that solves the whole day exactly; the agents decide when to call it, check its answer and explain the result (Section 4.4). Its earlier versions, without the optimiser tool and without retrieval, are design iterations: they were run on the exploratory benchmark and are reported in Section 5.3.1, but they are not compared systems. The pipeline is compared with the following alternatives:

- *No scheduling*: every appliance starts as soon as the user allows. It defines the saving of every other system.
- *Cheapest-hours automation*: the selection rule of the "Nordpool cheapest hours actions" blueprint for Home Assistant (Kert, n.d.). Within the appliance's window, the prices are sorted in ascending order and the threshold is the price at index $4h$ (counting from zero), where $h$ is the cycle length rounded up to whole hours. The blueprint switches a device on whenever the price is at or below the threshold;

for a cycle appliance, it is assumed that this trigger starts the cycle, which then runs to completion. With hourly prices, the same rule describes the scattered cheapest-hours mode of shelly-porssisahko (jisotalo, n.d.).

- *Cheapest window*: the block of whole hours, as long as the cycle rounded up, with the lowest average price between the ready time and the deadline. This is how the Nordpool Planner integration for Home Assistant (dala318, n.d.) and the sequential cheapest-hours mode of shelly-porssisahko (jisotalo, n.d.) choose when to switch a device on.

- *Smart timer (own heuristic)*: a peak-aware rule written for this study as a reference point, not a deployed product. It shows how far a simple rule gets without a language model or optimisation, and a household energy application could implement it. It places the appliances one at a time, the most energy-intensive first, each at its cheapest start at which the household load stays within the peak limit. The rule was fixed before it was run on the benchmark and was not tuned afterwards.

- *EMHASS* 0.18.4, the open-source optimizer Energy Management for Home Assistant (EMHASS contributors, 2026), run as households run it: as a Docker service with its default settings, called through its REST interface. Each scenario is sent as runtime parameters of a day-ahead optimisation: the prices, the base load as the load forecast, no photovoltaic production, the appliances as deferrable loads with their time windows and measured power profiles, and the peak limit as the maximum power drawn from the grid. Home Assistant itself is not involved, since all data is passed with the request. A configuration with a constant nominal power per appliance, set to the cycle's peak, is reported as a secondary result. As a cross-check, the same scenarios were also solved by calling EMHASS's optimisation module directly with an exact solver tolerance; the two routes give the same schedule in most scenarios, and the remaining differences stem from the default tolerance of the service.

- *LLM without tools*: the same model as the agents, given the day's appliance plan together with all household data and the price and measured base load of every slot in a single prompt. Its schedule is taken from one answer, without tools or a reasoning loop. This corresponds to the standard prompting baseline of Yao et al. (2023) and isolates the contribution of tool use.

The cheapest-hours rules plan each appliance on its own and ignore the household's total load, exactly as the real tools do. Because language model output varies between runs, each system that uses a language model is run three times per scenario, while the deterministic systems are run once.

The MILP solver of Section 4.5 is not a compared system. It confirms during benchmark construction that every scenario has a feasible schedule, it solves the days of the case library, and it is the solver behind the optimiser tool. Since a schedule cannot save more than the MILP schedule, its saving is reported as the upper bound in the result tables.

**Stages.** The system was developed on a first benchmark of 48 scenarios with hourly prices in six scenario types, over the design iterations of Section 5.3.1. A first confirmatory run, on 48 held-out scenarios with fifteen-minute prices (the Q set, Section 5.2), tested hypotheses registered in advance about the pipeline, the LLM without tools and an earlier single-agent design. Its results are reported in Appendix B and remain valid, but the thesis no longer builds its questions on the single-agent comparison.

The research design was then revised. The single agent was dropped as a compared system, the saving against no scheduling replaced the gap to the optimum as the primary outcome, retrieval and the optimiser tool were added, and the cheapest-window rule joined the existing tools. The six scenario types were replaced by a factorial design in which the price pattern, the household load and the peak limit vary separately (Section 5.1.4), because each of the old types changed several of these at once. The pre-registration records these changes as a dated amendment. Every system was then run on a new exploratory benchmark of 96 scenarios with this design. The hypotheses of Table 3.2 were fixed afterwards in a second pre-registration, committed to the project repository before any run on a second held-out benchmark of 96 scenarios (the Q2 set), whose days appear in no other benchmark and not in the case library. Only the run on Q2 is used to accept or reject them.

**Table 3.2 Hypotheses and statistical tests (second pre-registration, Q2 set).**

| | Hypothesis | Test |
|---|---|---|
| *SQ1, value and cost* | | |
| H1a | The Multi-Agent Pipeline saves more than the LLM without tools. | Wilcoxon signed-rank |
| H1b | Its recommendations are valid more often than those of the LLM without tools. | Wilcoxon signed-rank |
| H1c | Its mean response time is below 60 s and its mean API cost below \$0.10 per recommendation (two tests). | One-sided $t$ test against the limit |
| *SQ2, conditions (over all price patterns, loads and limits)* | | |
| H2a | The Multi-Agent Pipeline saves more than the cheapest window. | Wilcoxon signed-rank |
| H2b | It saves more than the cheapest-hours automation. | Wilcoxon signed-rank |
| H2c | It saves the same as EMHASS within $\pm 1$ percentage point. | Paired $t$ TOST |

### 3.3.1 Data collection

The study uses secondary data only; no new measurements were taken and no human participants were involved. Two kinds of input data are used. Appliance cycles, base loads and fixed appliances come from the REFIT Electrical Load Measurements dataset, which records appliance-level power in 20 UK households at eight-second resolution between 2013 and 2015 and is published under an open licence with the households anonymised (Murray et al., 2017). Day-ahead electricity prices for the Nord Pool FI bidding zone come from two public services that republish Nord Pool data, Sähkötin and Elering, and were

checked against the official prices of the ENTSO-E Transparency Platform (Section 5.1). Both are combined into benchmark scenarios by a fixed, seeded procedure, described in Section 5.1, so that every benchmark can be rebuilt exactly.

The evaluation data are generated by running the systems on these scenarios. Every run is stored as one record in a results file. The record contains the schedule, its cost, whether it is valid and which constraint it violates if not, the number of input and output tokens, the API cost, the response time, the number of tool calls and feedback rounds, and any error the agents reported. The reasoning trace and tool calls of every language model run are also recorded with LangSmith. EMHASS is called through its REST interface, and its schedules are scored by the same function as all other systems. Runs are resumable, so an interruption repeats only the run in progress, and a run that failed because of a provider or network error is retried and not recorded. Every completed run is kept, including invalid ones.

### 3.3.2 Data analysis

**Scoring**

All schedules are scored by the same function. The cost of a schedule is the consumer price of electricity multiplied by the energy of the flexible appliance cycles, summed over the day; the consumer price adds the retail margin, distribution fee, electricity tax and VAT to the spot price (Section 5.1). The base load costs the same under every schedule and is therefore excluded from the cost, but it counts towards the peak load limit. The peak limit is the highest total power the household may draw: in every fifteen-minute slot, the base load plus the power of the appliances running in that slot must stay at or below it. The benchmarks use two levels, a strict 3.5 kW and a realistic 8 kW, and treat the limit as a hard constraint rather than a priced one (Section 5.1). A schedule is valid when every appliance has a start time, every cycle runs within its user window, and the total household load never exceeds the peak limit. An invalid schedule, or a run that returns no schedule, is scored at the cost of no scheduling, on the reasoning that the household would then fall back to starting its appliances when they are ready. Starting every appliance when it is ready can itself exceed the limit, typically on heavy-load days under the strict limit; it nevertheless remains the reference cost, because it is what the household would do without a recommendation, so an invalid schedule saves nothing against it. The validity rate is reported separately, so that this convention does not hide failures.

The primary outcome is the saving against no scheduling:

$$\text{saving}(\%) = \frac{\text{cost without scheduling} - \text{system cost}}{\text{cost without scheduling}} \times 100.$$

It is what a household gains by following the recommendation, and it needs no optimum as a reference. Because the consumer price stays positive even when the spot price is negative, the cost without scheduling is always positive. Over a set of runs, the

pooled saving divides the summed savings by the summed costs without scheduling, so that expensive days weigh more than cheap ones, as on a bill; the paired tests use the per-scenario saving. An invalid run saves nothing. The gap to the MILP optimum, $(\text{system cost} - \text{MILP cost})/\text{MILP cost} \times 100$, is reported as a secondary measure, because it shows how much saving a system leaves on the table.

The choice of tariff does not change the ranking of the systems. The margin, distribution fee and electricity tax are constant per kilowatt-hour, and every cycle uses the same energy wherever it is placed, so the consumer price changes the cost of every schedule by the same amount and scales all cost differences by the common factor $1 + v$. The ranking of the systems and the euro differences between them are therefore the same as under the spot price, apart from that factor, which was also verified on the recorded schedules. The saving in percent does depend on the tariff, through its denominator: under the spot price alone, the same euro saving would be a larger share. The consumer price is used because it is what the household pays.

The sub-questions use these measures as follows. SQ1 compares the saving and validity of the pipeline with those of the LLM without tools, and checks its response time and API cost per recommendation against the design objectives of Section 4.1 (60 s and \$0.10); tokens (input plus output) are reported alongside. SQ2 compares the saving and validity of the pipeline with those of the existing home energy management tools, the cheapest-hours rules and EMHASS, first over all scenarios and then separately for each price pattern, household load and peak limit. The breakdown by condition is descriptive; the registered tests concern all scenarios together. Also reported without hypotheses are the total euro saving, the smart timer (own heuristic), the gap to the optimum, and the conflict resolution rate, which is the share of valid runs among the scenarios that require coordination (those in which starting every appliance at its individually cheapest time would exceed the peak limit).

**Statistical analysis**

Two systems are compared per scenario, using the mean saving over the runs as the paired observation. The Wilcoxon signed-rank test assesses whether the difference across the 96 paired scenarios is statistically significant. It is designed for paired comparisons and does not assume that the differences are normally distributed. Pratt's treatment of zero differences is used, so that scenarios in which both systems reach the same cost remain in the ranking instead of being discarded. Practical significance is reported with the effect size $r = |Z|/\sqrt{N}$. All tests are two-sided at $\alpha = 0.05$. A directional hypothesis is supported when its test is significant and the effect has the predicted direction. Two-sided tests are the conservative choice for directional hypotheses, since a one-sided test would halve each $p$-value. The $p$-values of the registered hypotheses are adjusted for multiple testing with the Holm procedure within each sub-question: four tests for SQ1 and three for SQ2. The two tests of H1c compare a mean with a fixed limit, which is a one-sided claim by nature, so they are one-sided: the upper end of the one-sided 95 % confidence interval must lie below the limit.

A non-significant difference does not by itself show that two systems perform alike, since it can also result from too little data. Equivalence with EMHASS (H2c) is therefore tested with the two one-sided tests (TOST) procedure (Schuirmann, 1987; Lakens, 2017). The systems are considered equivalent when both one-sided tests reject, at the 5 % level, a mean difference in saving of at least one margin in either direction. This is the same as requiring the 90 % confidence interval of the mean paired difference to lie inside the margin. The margin is $\pm 1$ percentage point of the cost without scheduling. On the Q2 set that cost has a median of €0.61 per day (interquartile range €0.28 to €1.06, maximum €2.65), so one percentage point is about 0.6 cents on the median day and under three cents on the most expensive one, which a household would not notice on its bill. The paired $t$ version of the test concerns the mean difference and is the primary one; a Wilcoxon version, which concerns the typical scenario, and margins of 0.5 and 2 percentage points are reported alongside.

# 4. System Design and Implementation

This chapter describes the design and implementation of the hierarchical ReAct multi-agent system, the Multi-Agent Pipeline. It presents the problem definition and design objectives, the architecture of the pipeline, the specifications of each agent, the retrieval of solved days and the optimiser tool, the MILP solver, and the technology stack used throughout.

## 4.1 Problem definition and objectives

Based on the literature synthesis in Section 2.8 and DSR Phase 2, the artifact must satisfy five design objectives. First, multi-appliance coordination: the pipeline must correctly schedule two or more flexible appliances simultaneously without producing overlapping high-load windows. Second, price-volatility robustness: the pipeline must produce cost-reducing schedules under every price pattern, household load and peak limit of the benchmark. Third, constraint compliance: every schedule must respect user-defined completion deadlines and household load limits. Fourth, interpretability: the full reasoning trace of every scheduling decision must be captured through LangSmith. Fifth, computational efficiency: the pipeline must complete scheduling within 60 seconds and under $0.10 API cost per scenario using GPT-4o-mini.

## 4.2 Architecture overview

The artifact is a LangGraph state graph in which each node is either a ReAct agent or a coordination step. All nodes read from and write to a shared state object that persists for the whole run (Table 4.1). The Orchestrator first turns the household's request into a task brief listing the appliances to schedule. The Appliance Profiler then classifies the appliances and records the power draw, cycle duration and time window of every flexible one. The Price-Aware Scheduler proposes a start time for each flexible appliance from the price curve, and the Conflict Resolver checks the proposal against the household peak load limit and staggers start times where appliances overlap.

After the Resolver, the Orchestrator's verification step checks the resolved schedule for three kinds of problems: remaining peak load conflicts, detected with the same conflict tool that the Resolver uses; requested appliances that have no start time; and start times outside the user's time window. If a problem remains, the step sends a description of it back to the Scheduler, which proposes a revised schedule for the Resolver to check again. The Scheduler and Resolver run at most twice, that is, with one feedback round. The schedule is then finalised, and the Orchestrator writes a short summary for the household that states the result of the final peak load check. The original request travels in the shared

state together with the Orchestrator's brief, so an appliance missing from the brief can still be profiled and scheduled. Figure 4.1 shows the architecture, and Table 4.1 lists the fields of the shared state.

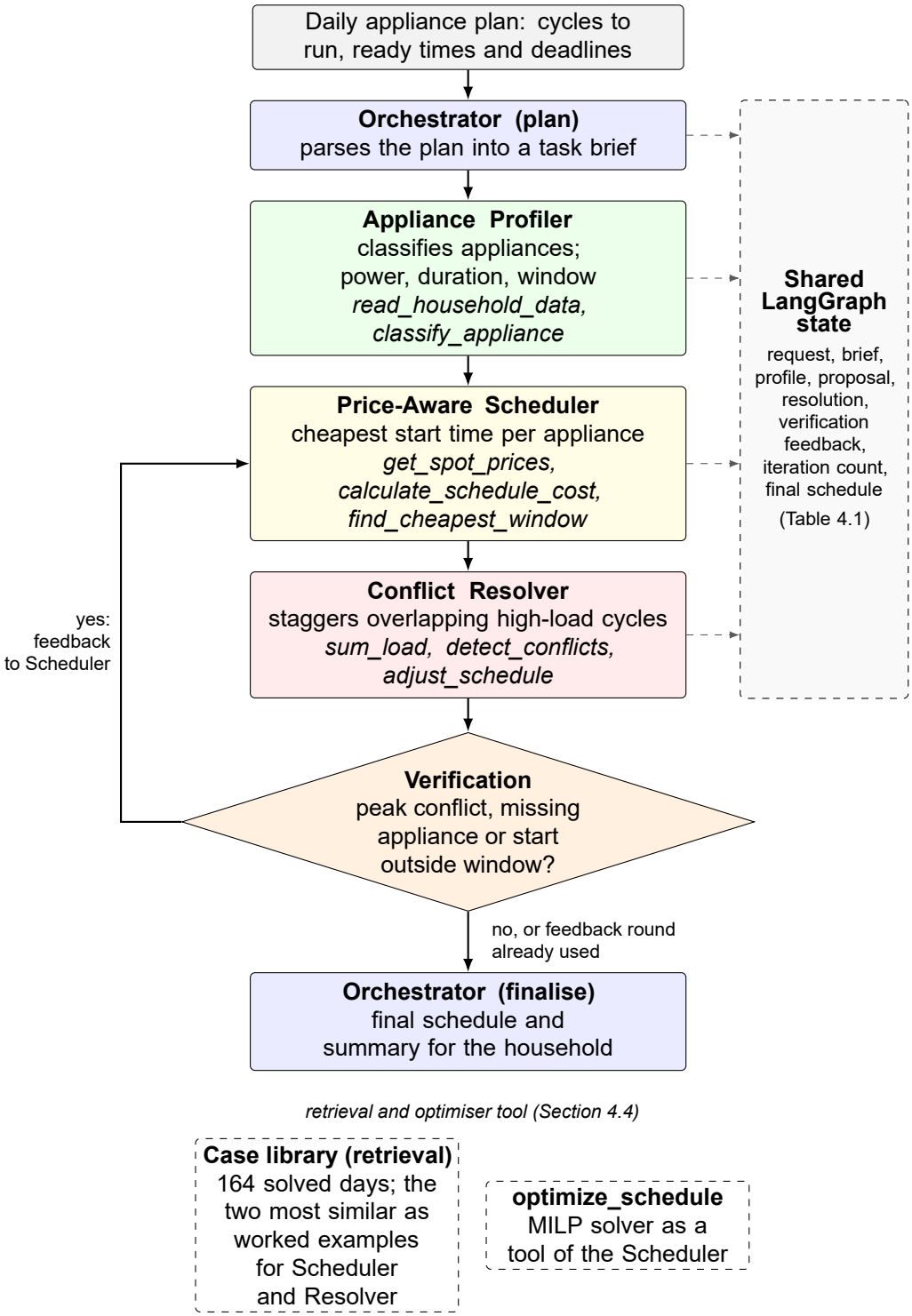



**Figure 4.1. Architecture of the hierarchical ReAct multi-agent system.**

**Table 4.1 Fields of the shared LangGraph state.**

| Field | Written by | Content |
|---|---|---|
| request | caller | the household's scheduling request in natural language |
| brief | Orchestrator | appliance identifiers to schedule and stated constraints |
| profile | Profiler | per appliance: type, flexible or fixed, peak power, cycle duration, earliest start and deadline |
| proposal | Scheduler | start time per appliance, estimated cost and a short justification |
| resolution | Resolver | conflict-free start times, whether conflicts remain, and the changes made |
| verification feedback | verification step | remaining peak conflicts, appliances without a start time and start times outside the user window; empty when the schedule passes |
| iterations | verification step | number of Scheduler–Resolver passes completed (at most two) |
| final, summary | Orchestrator | final start times and the summary shown to the household |
| tool calls, recoveries, errors | all nodes | counters and error messages recorded for the evaluation |

## 4.3 Agent specifications

The **Orchestrator** receives the household scheduling request, coordinates the three specialist agents in sequence, and writes the final output. It does no scheduling itself, so its context stays small.

The **Appliance Profiler** reads the household consumption data and classifies each appliance by shiftability. The classification distinguishes flexible loads, which have long cycle durations and user-defined completion windows (washing machines, dishwashers and tumble dryers), from fixed loads such as refrigerators and microwaves. The Profiler outputs a structured appliance profile containing power draw in Watts, typical cycle duration in minutes, and the constraint window for each flexible load.

The **Price-Aware Scheduler** receives the appliance profile and the day-ahead prices. It finds the cheapest window for each flexible appliance within its user window and proposes a daily schedule with a start time per appliance and an estimate of the total cost.

The **Conflict Resolver** receives the proposed schedule and looks for times when two or more appliances with high power draw would run together and exceed the household's peak load limit. If it finds such a conflict, it staggers start times to remove it, choosing the moves that add the least to the total cost.

Each specialist is a ReAct agent that may only call its own tools and must end with a

structured output (the appliance profile, the proposed schedule or the resolved schedule), which is validated against a schema before it is written to the shared state. The agents share a budget of 75 graph steps, divided by workload: 15 for the Profiler, 35 for the Scheduler and 25 for the Resolver. The Scheduler received the largest share after the pilot runs showed that it performs most of the tool calls. When a model replies in prose instead of producing the required structured output, the agent is resumed from its last state and asked once more to return the output, at most twice. This rule applies identically to every agent and model, and every such recovery is recorded.

Table 4.2 lists the tools. They are deterministic functions over the scenario data, so every number an agent reasons with comes from a calculation rather than from the language model. The first seven tools each handle part of the problem: `find_cheapest_window`, for example, considers one appliance at a time and ignores both the other appliances and the peak load limit. With these tools alone, coordinating the appliances is left to the agents. The eighth, `optimize_schedule`, solves the whole day and was added in the last design iteration (Section 4.4).

Before the optimiser tool was added, the automatic mode of `adjust_schedule` was the agents' only way to repair a conflict. It optimizes the start of one appliance while all others stay where they are and never re-plans several appliances together. Which appliances to move, in which order, and when to stop were decisions of the agent, and a poor sequence of single moves could settle on a schedule far from the optimum. In scenario D06 of the first benchmark, for example, the pipeline ended about 30 % above the optimal cost with a staggered schedule that no single further move could improve. This limitation led to the retrieval of solved days and the optimiser tool described next.

**Table 4.2 Tools available to the agents.**

| Tool (inputs) | Agent | Output and logic |
|---|---|---|
| `read_household_data()` | Profiler | Peak load limit, measured base load range, and for each appliance its type, cycle duration, energy, peak power and user window (flexible) or rated power (fixed). |
| `classify_appliance(appliance_id)` | Profiler | Flexible or fixed, with a reason; washing machines, dishwashers and tumble dryers are shiftable cycle loads. |
| `get_spot_prices(from_time, to_time)` | Scheduler | Consumer price in c/kWh for every 15-minute slot, optionally for a time range. |
| `calculate_schedule_cost(appliance_id, start_time)` | Scheduler | Cost of one cycle started at the given time (power profile $\times$ price $\times$ 0.25 h, summed over the cycle), its end time, and whether it lies within the user window. |
| `find_cheapest_window(appliance_id, top_k)` | Scheduler | The $k$ cheapest start times within the user window for one appliance, found by evaluating every valid start. |
| `sum_load(schedule)` | Resolver | Maximum total household load (base load plus scheduled cycles) and the five highest slots, compared with the limit. |
| `detect_conflicts(schedule)` | Resolver | Time ranges in which the total load exceeds the limit and the appliances running then, or confirmation that there is no conflict. |
| `adjust_schedule(schedule, appliance_id, new_start_time)` | Resolver | Moves one appliance and reports the cost change, the window check and any remaining overload. Without a start time, it moves the appliance to its cheapest start at which it causes no overload, keeping the others fixed, and lists alternatives. |
| `optimize_schedule(appliance_ids)` | Scheduler* | Solves the day for the given appliances with the MILP of Section 4.5 and returns the start times, the cost and the maximum load. *Added in the last design iteration. |

## 4.4 Retrieval of solved days and the optimiser tool

**Case library.** Retrieval gives the agents examples of how similar days were solved best. The case library holds 320 cases: 20 price days for each combination of price pattern and household load, each under both peak limits. The days are drawn from the hourly prices up to 30 September 2025, excluding every day of every benchmark, so no evaluated day is in the library. Each case is composed exactly like a benchmark scenario (Section 5.1), with the same REFIT cycle pool, price patterns, loads and limits, and solved with the MILP of Section 4.5. A deployed system would build such a library from the household's own past days in the same way.

**Retrieval.** Each day is described by two vectors: the shape of its price curve (24 hourly consumer prices, standardised to mean zero and unit variance) and six structural values (the number of appliances, their combined peak power relative to the headroom under the limit, the mean window length, the share of appliance pairs whose windows overlap, whether the individually cheapest starts would exceed the limit, and the relative price spread). The similarity of two days is the mean of the cosine similarity of the price shapes and $e^{-2d}$, where $d$ is the Euclidean distance between the structural values. The two most similar cases are retrieved. The description and the weights were set once, before any evaluation run, and were not tuned. The retrieval is deterministic and needs no embedding model.

**Worked examples.** Each retrieved case is turned into a short text in the agents' own vocabulary: the number of appliances, the peak limit and the price range; whether the individually cheapest starts would exceed the limit; the optimal start of every appliance with its own cheapest start and the extra cost of the difference; and the pattern of the staggering, for example that the tumble dryer starts as the washing machine ends. The text is appended to the messages of the Scheduler and the Resolver with an instruction to use the examples as patterns, not as answers, and to check the day's own prices and loads with the tools.

**Optimiser tool.** The Scheduler also has `optimize_schedule` (Table 4.2), and its prompt asks it to call the tool once with all flexible appliances, to check any unusual start time with `calculate_schedule_cost`, and to fall back to `find_cheapest_window` if the solver finds no schedule. The Resolver and the verification step are unchanged, so every schedule is still checked independently of the tool that produced it, as Jonah et al. (2026) recommend. This follows the division of work of Li et al. (2023): the solver searches, and the agents formulate the call, check the result and explain it to the household.

The pipeline reached this form over several design iterations (Section 5.3.1): the base pipeline without retrieval or the optimiser tool, then the pipeline with retrieval only. Both earlier versions were run on the exploratory benchmark, and their results show what each addition contributed. An earlier version of the study also compared the base pipeline with a single ReAct agent that had the first seven tools of Table 4.2 and the same step budget; that comparison is summarised in Section 5.4. Only the final Multi-Agent Pipeline is a compared system in the current evaluation.

## 4.5 MILP solver

The MILP solver computes the optimal schedule of a day. It is used in three roles: it confirms during benchmark construction that every scenario has a feasible schedule, it solves the days of the case library, and it is the solver behind the optimiser tool. Its schedules also give the upper bound on the saving in the result tables. The model is

the standard time-indexed formulation of appliance scheduling with binary start variables (Sou et al., 2011). The day is divided into slots $t \in \mathcal{T} = \{0, \dots, 95\}$ of $\Delta = 0.25\,\text{h}$. Each flexible appliance $a \in \mathcal{A}$ has a power profile $p_{a,0}, \dots, p_{a,d_a-1}$ in kW over its cycle of $d_a$ slots, an earliest start slot $e_a$ and a deadline slot $\ell_a$ by which the cycle must end. Its admissible start slots are

$$\mathcal{S}_a = \{\, s \in \mathbb{Z} : e_a \le s \le \ell_a - d_a \,\}.$$

The model uses binary start variables $x_{a,s} \in \{0, 1\}$, equal to one if appliance $a$ starts in slot $s \in \mathcal{S}_a$. With the consumer price $\pi_t$ in €/kWh, the base load $b_t$ in kW and the household peak limit $\bar{P}$ in kW, the model is

$$\min_{x} \quad \sum_{a \in \mathcal{A}} \sum_{s \in \mathcal{S}_a} c_{a,s}\, x_{a,s}, \qquad c_{a,s} = \Delta \sum_{k=0}^{d_a-1} p_{a,k}\, \pi_{s+k}, \tag{4.1}$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}_a} x_{a,s} = 1 \qquad \forall a \in \mathcal{A}, \tag{4.2}$$

$$b_t + \sum_{a \in \mathcal{A}} \sum_{\substack{s \in \mathcal{S}_a \\ 0 \le t-s < d_a}} p_{a,t-s}\, x_{a,s} \le \bar{P} \qquad \forall t \in \mathcal{T}, \tag{4.3}$$

$$x_{a,s} \in \{0, 1\} \qquad \forall a \in \mathcal{A},\ s \in \mathcal{S}_a.$$

The objective (4.1) is the consumer cost of the flexible cycles, with $c_{a,s}$ the precomputed cost of starting appliance $a$ in slot $s$. Constraint (4.2) requires exactly one cycle per appliance, and constraint (4.3) keeps the total household load within the peak limit in every slot. Because a decision fixes the start of a whole cycle, each cycle runs without interruption once started, and restricting the variables to $\mathcal{S}_a$ keeps every cycle within its user window, so neither property needs a separate constraint. The model is solved with PuLP and the CBC solver in well under a second per scenario. Its correctness was verified against exhaustive enumeration of all start combinations on small scenarios.

## 4.6 Technology stack

Table 4.3 lists the software used to build and evaluate the artifact, with the versions used for the reported results. All components except the commercial language model are open source. The API cost of the commercial model is an outcome of the evaluation and is reported with the results in Chapter 5.

**Table 4.3 Technology stack.**

| Component | Tool and version | Purpose |
|---|---|---|
| Programming language | Python 3.12 | implementation of the artifact, baselines and analysis |
| Agent orchestration | LangGraph 1.2, LangChain 1.4 | state graph of the pipeline, ReAct agents, tool calling and schema-validated outputs |
| Commercial LLM | GPT-4o-mini (snapshot 2024-07-18), OpenAI API | model of the Multi-Agent Pipeline, its design iterations and the LLM without tools |
| Open-weight LLM | gpt-oss-20b, Ollama 0.34 and 0.35 | open-weight model hosted on local hardware, used in an exploratory run of the earlier design only; Ollama updated itself from 0.34.4 to 0.35.0 during that run |
| MILP solver | PuLP 3.3 with CBC | benchmark feasibility check, case library and the optimiser tool |
| HEMS optimizer | EMHASS 0.18.4, Docker | open-source home energy management optimizer compared in SQ2, called through its REST interface |
| Data processing | pandas 3.0, NumPy 2.5 | REFIT cycle extraction, price series and scenario construction |
| Tracing | LangSmith | recording the reasoning trace and tool calls of every run |
| Statistics | SciPy 1.18 | Wilcoxon signed-rank test, TOST equivalence test, effect size and Spearman correlation |

# 5. Evaluation and Results

This chapter presents the evaluation of the artifact, corresponding to DSR Phases 4 and 5. Section 5.1 describes the benchmarks: the appliance and price data, the first benchmark of 48 scenarios in six types, and the factorial benchmarks of 96 scenarios used for the final design. Section 5.2 uses the MILP solver alone to examine Finland's move to fifteen-minute electricity prices, quantifying how much of the achievable scheduling saving depends on fifteen-minute rather than hourly price information; this establishes where differences between schedulers can be expected before any agent is evaluated. Section 5.3 reports the demonstration of the pipeline on representative scenarios (DSR Phase 4) and the design iterations that led to it. Section 5.4 summarises the evaluation of the earlier design, which compared the pipeline with a single agent and is reported in full in Appendix B. Section 5.5 follows the final design through its last iterations on the exploratory factorial benchmark, and Section 5.6 reports the pre-registered test of the final Multi-Agent Pipeline on the held-out set Q2, which answers SQ1 and SQ2. Section 5.7 reports a design iteration made after that test, which halves the pipeline's response time.

## 5.1 Dataset and scenario design

The evaluation benchmark combines real appliance consumption profiles from UK households with real Finnish day-ahead electricity prices. Each of the 48 scenarios describes one day in 96 fifteen-minute slots (00:00–24:00 local time) and consists of a price curve, a household base load, a set of flexible appliance cycles with user time windows, and a household peak load limit. The benchmark is generated by a deterministic, seeded build procedure, so it can be reproduced exactly from the source data.

### 5.1.1 Appliance data

Appliance profiles are drawn from the cleaned REFIT Electrical Load Measurements dataset (Murray et al., 2017; Murray & Stankovic, 2016), which records the whole-house aggregate and nine individually monitored appliances at 8-second intervals in 20 UK households between 2013 and 2015. The cleaned release corrects timestamps for daylight saving time and re-aligns appliance monitor columns where a monitor was moved between appliances during the study. The appliance-to-column mapping follows the dataset documentation.

Eighteen households are used. House 12 is excluded because it has no monitored washing machine, dishwasher or tumble dryer. House 19 is excluded after inspection: its washing-machine monitor recorded no heating phase in any of its 246 detected cycles (no fifteen-minute mean above 1.2 kW, against 22–70 % of cycles in the other households),

so its profiles do not represent the water-heating load that dominates the cost and peak demand of a wash. Washer-dryers (houses 1, 8, 9 and 18) are not used as profiles, because they run washing and drying as a single cycle; their consumption is nevertheless removed from the base load.

Individual appliance cycles are extracted from each monitor by thresholding the one-minute mean power at 20 W and merging pauses shorter than 30 minutes for dishwashers (whose programmes contain a passive drying pause) and 15 minutes for washing machines and tumble dryers (which are often reloaded within minutes). Cycles shorter than 20 minutes, longer than 300 minutes or below 0.1 kWh are discarded, and each remaining cycle is converted into a fifteen-minute mean power profile. This yields 6,172 washing-machine cycles from 19 monitored machines, 4,702 dishwasher cycles from 15 machines and 1,940 tumble-dryer cycles from 7 machines. To prevent heavily used appliances from dominating the benchmark, a scenario first draws a monitored appliance of the required type and then one of its cycles. The household base load is the median, per fifteen-minute slot of the day, of the aggregate minus all shiftable appliance monitors, computed after discarding readings that the dataset flags as inconsistent (sub-metered sum exceeding the aggregate).

### 5.1.2 Electricity price data

Day-ahead spot prices for the Nord Pool FI bidding zone come from two public services that republish Nord Pool data. The hourly prices of the exploratory benchmark and the case library, 1 January 2022 to 30 September 2025, come from Sähkötin (Pakastin Oy, 2026); each hourly value is repeated across the four fifteen-minute slots of its hour. The fifteen-minute prices of the held-out benchmarks, from 1 October 2025, come from Elering (Elering AS, 2026). All values are in EUR/MWh excluding VAT, and days with a daylight-saving transition are dropped. Both sources were checked against the official day-ahead prices of the ENTSO-E Transparency Platform for the whole period (ENTSO-E, 2026). The Elering prices are identical to the official ones at all 67,411 timestamps. The Sähkötin prices are identical for every hour before October 2025 except one hour on a daylight-saving transition day, which no benchmark uses. Every price in the scenarios of the exploratory and the held-out benchmarks therefore matches the official data.

Scheduling cost is computed with the consumer price rather than the raw spot price. For spot price $s$, the consumer price per kWh is

$$p = (s + m + d + \tau)(1 + v),$$

with a retail margin $m = 0.5$ c/kWh, a distribution fee $d = 5.0$ c/kWh, the electricity tax $\tau = 2.253$ c/kWh and VAT $v = 25.5\,\%$. These are representative values for a Finnish household on a spot-price contract. The tax is that of tax class I including the security-of-supply fee, valid until 31 March 2026; on 1 April 2026 the fee rose from 0.013 to 0.085 c/kWh, raising the tax to 2.325 c/kWh (Caruna, 2026). The benchmark keeps one tariff for all days, including a single VAT rate, so that scenarios from different

periods remain comparable. Because every component is constant per kilowatt-hour, the choice of values does not change the optimal schedule, the ranking of the systems or the savings captured, only the size of the optimality gap (Section 3.3). Using the consumer price keeps costs positive on days with negative spot prices, which keeps the optimality gap defined in Section 3.3 meaningful. Only the energy of the flexible appliances is costed, since the base load costs the same under every schedule.

### 5.1.3 Scenario construction

Table 5.1 summarises the six scenario types. Days are assigned to types by the within-day standard deviation of the spot price. Types A and B use disjoint days from the lowest quartile among days without negative prices, type C uses days between the 40th and 60th percentiles, and type D uses days in the top 15 % that fall in December–February. Type E uses days from December 2022, sampled from the 16 days with the highest maximum price. Type F uses days with at least two hours of negative prices. Eight days are sampled per type.

Types C–F need up to five loads, while REFIT provides three flexible appliance types; a scenario may therefore contain two cycles of the same type, such as two washing loads. Each appliance has a user window (earliest start and deadline), and its naive start time is the earliest start. Types A–C use windows opening between 06:00 and 12:00 and lasting 8–14 hours. In type D all five appliances share one window from 08:00–10:00 until 22:00–24:00, so their cheapest start times coincide. In type E each appliance becomes ready 0–2 hours before the day's highest daytime price and has 4–8 hours of slack beyond its cycle, so the naive schedule runs into the price spike while avoiding it remains possible. This reflects that price spikes coincide with evening demand peaks, when households tend to start appliances. Type F windows open between 00:00 and 04:00 and last until midnight.

The household peak load limit, including the base load, is 3.5 kW in every fifteen-minute slot. It is a deliberately strict, stylized cap that makes the appliances compete for capacity, rather than a limit that Finnish households face today. So that every appliance can at least run on its own, the limit of a scenario is raised to the day's highest base load plus the highest power of its largest cycle plus 0.2 kW where that sum exceeds 3.5 kW, and in further steps of 0.5 kW if the optimal schedule is still infeasible. In the factorial sets this applies to 6 (exploratory main set) and 8 (Q2) of the 48 scenarios under the strict limit, with limits of up to 5.75 kW. The main fuse (typically $3 \times 25$ A, about 17 kW) would never bind for these appliances. Power-based distribution tariffs exist, but they price the peak rather than forbid it. Helen Sähköverkko's time-based distribution product, for example, charges € 1.57 per kW and month, excluding VAT, for the third-highest hourly average power of the month, with 80 % of night-time power counted (Helen Sähköverkko, 2026a). Under the Energy Authority's guidance, which network companies must follow by 1 January 2029 at the latest, a household power charge is based on the highest 60-minute average power of the month, starts only above 8 kW, and remains optional, since a product without a power charge must still be offered (Helen Sähköverkko, 2026b). Compared with these tariffs, the benchmark's cap is stricter in level (3.5 instead of 8 kW), in time resolution

(15 instead of 60 minutes) and in kind (a hard limit instead of a price). It therefore tests coordination under demanding conditions; replacing it with a priced peak is left for future work. The build procedure verifies with the MILP solver that every scenario has a feasible schedule; this held for all 48 scenarios without relaxing the limit. Table 5.2 reports the resulting characteristics per type. A scenario requires coordination when running every appliance at its individually cheapest start time exceeds the peak limit; these 28 scenarios form the basis of the conflict resolution rate.

**Table 5.1 Evaluation scenario design.**

| Type | Price pattern | Appliances | Inst. | Research purpose |
|---|---|---|---|---|
| A | Stable, low volatility | 2 appliances: dishwasher + washing machine | 8 | Baseline check: both systems should perform well |
| B | Stable, low volatility | 3 appliances: dishwasher + washing machine + tumble dryer | 8 | Tests three-way conflict resolution |
| C | Moderate volatility | 3 to 5 appliances mixed | 8 | Tests price-signal responsiveness |
| D | High volatility, winter peak | 5 appliances, overlapping constraints | 8 | Most financially consequential condition |
| E | Price spike (Dec 2022 crisis) | 3 appliances, ready 0–2 h before the daily price peak, 4–8 h slack | 8 | Extreme Finnish price event robustness |
| F | Negative spot price (surplus) | 4 appliances, flexible windows | 8 | Tests use of low-cost windows on negative spot-price days |

**Table 5.2 Benchmark characteristics per scenario type.**

| Type | Price days | Mean appliances | Require coordination | Naive schedule within limit | MILP saving vs. naive |
|---|---|---|---|---|---|
| A | 2022–2026 | 2.0 | 1/8 | 8/8 | 10.2 % |
| B | 2023–2026 | 3.0 | 4/8 | 7/8 | 4.3 % |
| C | 2023–2026 | 3.4 | 5/8 | 6/8 | 19.9 % |
| D | 2022–2026 (Dec–Feb) | 5.0 | 8/8 | 0/8 | 37.0 % |
| E | 1–14 Dec 2022 | 3.0 | 4/8 | 3/8 | 14.4 % |
| F | 2023–2025 | 4.0 | 6/8 | 5/8 | 7.8 % |
| All | | 3.4 | 28/48 | 29/48 | |

### 5.1.4 Factorial benchmarks

The six types above change several properties at once: type D, for example, differs from type B in its price pattern, its number of appliances and its window structure, so an effect on type D cannot be attributed to one of them. The benchmarks used for the final design therefore follow a factorial design in which three factors vary separately. The *price pattern* has four levels: stable (the lowest quarter of days by within-day standard deviation), volatile (the top 15 %), spike (the days with the highest daily maximum) and negative (at least one hour below zero). The *household load* has two levels: light, a dishwasher and a washing machine with separate daytime windows, and heavy, two washing loads, two drying loads and a dishwasher that share one window. The *peak limit* has two levels: strict, the 3.5 kW cap described above, and realistic, 8 kW, the threshold above which the Energy Authority's guidance allows a power charge. Six price days are drawn for each combination of price pattern and load, and each day is scheduled under both limits with the same appliances, so the effect of the limit is measured on identical days. A benchmark therefore has 48 days and 96 scenarios.

Two benchmarks were built with this design, and no day appears in both or in any earlier benchmark. The exploratory main set uses the hourly prices up to 30 September 2025. The held-out set Q2 uses the native fifteen-minute prices from 1 October 2025 to 25 September 2026, and was built before any system was run on it. The case library for retrieval (Section 4.4) is built with the same design from 320 further days up to 30 September 2025. In Q2, 30 of the 96 scenarios require coordination, almost all of them heavy-load days under the strict limit, and every scenario has a feasible schedule. The cost of starting every appliance when it is ready has a median of €0.56 per scenario (interquartile range €0.29 to €0.91, maximum €2.81).

## 5.2 Value of 15-minute price information

Since 1 October 2025 the Nord Pool day-ahead market settles Finnish electricity in fifteen-minute periods. The main benchmark uses hourly prices, because its type E days date from the 2022 energy crisis. To examine the new market design, a second set of 48 scenarios (Q-A01 to Q-F08) was built with the same procedure from native fifteen-minute FI prices for 1 October 2025 to 25 September 2026 (358 complete days), obtained from the public price service of the Estonian transmission system operator, which republishes Nord Pool data (Elering AS, 2026). Its hourly averages match the hourly series used in Section 5.1. Because this period contains no crisis month, its type E days are drawn from the 16 days with the highest peak price (up to 65 c/kWh), and no day is used by more than one type. Prices genuinely vary within the hour: in 98 % of hours the four quarter-hour prices differ, with a median spread of 0.75 c/kWh.

For each Q scenario, two optimal schedules were computed with the MILP solver. The first is planned on the real fifteen-minute prices. The second is planned on hourly averages, which is all the price information a household had before the switch, and is then settled at the real fifteen-minute prices. Hourly prices make the four quarters of an hour tie, so

ties are broken towards the earliest start. The cost difference between the two schedules is the saving that is only reachable with fifteen-minute price information. Table 5.3 reports the savings of both schedules relative to the naive schedule, the additional cost reduction from fifteen-minute information, and the share of the achievable saving that requires it; the "All" row weights scenarios by their cost.

**Table 5.3 Optimal schedule cost with hourly vs. 15-minute price information (MILP, Q scenarios).**

| Type | Intra-hour spread (c/kWh) | Saving, hourly plan | Saving, 15-min plan | Additional cost reduction | Share of saving needing 15-min | Scenarios improved |
|---|---|---|---|---|---|---|
| A | 0.41 | 4.1 % | 4.2 % | 0.1 % | 2.4 % | 5/8 |
| B | 0.33 | 3.2 % | 3.4 % | 0.3 % | 6.4 % | 8/8 |
| C | 1.37 | 7.1 % | 8.6 % | 1.7 % | 16.4 % | 8/8 |
| D | 4.96 | 27.6 % | 28.6 % | 1.4 % | 3.5 % | 8/8 |
| E | 4.91 | 35.4 % | 37.0 % | 2.1 % | 2.8 % | 6/8 |
| F | 0.21 | 4.6 % | 4.7 % | 0.1 % | 1.7 % | 4/8 |
| All | 2.03 | 23.7 % | 24.7 % | 1.2 % | 3.7 % | 39/48 |

Fifteen-minute price information lowers the optimal cost in 39 of the 48 scenarios, but the effect is modest: 1.2 % overall, or 3.7 % of the achievable saving. The remaining 96 % of the saving is available to a scheduler that plans on hourly prices, since most of the value lies in moving appliance cycles between hours rather than between quarters of an hour. The additional reduction grows with the intra-hour price spread (Spearman $\rho = 0.66$ across scenarios) and is largest on volatile days: 1.4 % in type D, 2.1 % in type E, and 10.2 % in the single largest case (Q-E03, 17 October 2025). Type C shows the largest relative share (16.4 %), because its hourly price differences are small, so intra-hour variation forms a larger part of the opportunity. In absolute terms the gain is about one cent per household per day for the two to five cycles scheduled, and about three cents on type D days. The schedules nevertheless differ: 110 of the 163 appliance start times change between the two plans. These results indicate that the hourly main benchmark captures most of the scheduling problem, while any difference between the agents under fifteen-minute prices is expected to appear mainly in types C, D and E.

## 5.3 Demonstration results

Table 5.4 shows what each system recommends for one day of the held-out set Q2, scenario Q2-PH01-35 (15 October 2025). It is a spike day with the heavy household load under the strict limit. Two washing machine cycles, two tumble dryer cycles and a dishwasher are all ready at 08:45 and must finish by midnight, and the household's total load, including a base load of 0.2 to 0.8 kW, must stay within 3.58 kW, the strict limit raised so that the largest cycle fits on top of the base load (Section 5.1). The consumer price peaks at 46 c/kWh at

09:00, just after the appliances are ready, and falls to 13 c/kWh at 21:00 and 10 c/kWh by midnight. Starting every cycle when it is ready would cost €2.57 and draw 7.0 kW, twice the limit.

**Table 5.4 Daily recommendations for scenario Q2-PH01-35 (15 October 2025, spike price pattern, heavy load, limit 3.58 kW; run 1 of the systems with a language model, GPT-4o-mini). Saving in percent of the cost without scheduling, €2.57; an invalid schedule saves nothing.**

| System | Washing 1 | Washing 2 | Drying 1 | Drying 2 | Dishwasher | Peak (kW) | Saving |
|---|---|---|---|---|---|---|---|
| No scheduling | 08:45 | 08:45 | 08:45 | 08:45 | 08:45 | 7.04 | – |
| Multi-Agent Pipeline | 19:00 | 22:45 | 21:15 | 21:45 | 22:30 | 3.54 | 70.2 % |
| EMHASS, measured profiles | 19:00 | 22:45 | 22:00 | 21:15 | 22:30 | 3.55 | 70.2 % |
| Smart timer (own heuristic) | 19:00 | 22:45 | 22:00 | 21:15 | 22:30 | 3.55 | 70.2 % |
| EMHASS, nominal power | 19:00 | 19:45 | 21:00 | 22:30 | 17:00 | 3.27 | 60.6 % |
| Cheapest window | 19:00 | 22:00 | 22:00 | 22:00 | 22:00 | 6.36 | invalid |
| Cheapest-hours automation | 11:45 | 20:45 | 20:30 | 20:30 | 20:30 | 5.34 | invalid |
| LLM without tools | 08:45 | 09:00 | 10:15 | 11:45 | 13:15 | 2.93 | 23.8 % |
| Optimum (upper bound) | 19:00 | 22:45 | 21:15 | 21:45 | 22:30 | 3.54 | 70.2 % |

The Multi-Agent Pipeline returned the optimal schedule. It moves every cycle into the evening and staggers the starts between 19:00 and 22:45, so that the load peaks at 3.54 kW, and saves €1.80, or 70.2 % of the cost without scheduling. EMHASS with measured profiles and the smart timer found schedules of the same cost that differ only in the order of the two dryers. The cheapest-hours tools found the same cheap evening hours but planned each appliance on its own. The cheapest window starts the second washing machine, both dryers and the dishwasher together at 22:00, and the cheapest-hours automation starts three of them at 20:30, so their load reaches 6.4 and 5.3 kW. The cheapest window's schedule would have cost almost exactly as little as the optimum, but it breaks the limit and therefore counts as saving nothing. On the same day under the realistic limit of 8 kW (scenario Q2-PH01-80), the two rules are valid and save 70.2 and 60.0 %. EMHASS configured with a constant nominal power plans every cycle at its peak power, which leaves too little room in the evening, and it moved the dishwasher to 17:00, saving 60.6 %.

The systems with a language model also explain their recommendation. The pipeline's Orchestrator wrote that "washing machine 1 will run at 19:00, followed by washing machine 2 at 22:45. The tumble dryers will operate at 21:15 and 21:45, while the dishwasher will run at 22:30", with a maximum load "comfortably within your limit of 3.58 kW". The summary is written from the result of the final peak-load check (Section 4.2), so its start times and its statement about the limit are correct. It does not, however, state the cost or the saving and calls the cost only "efficient". The LLM without tools kept every cycle between 08:45 and 13:15, the most expensive part of the day, and saved 23.8 % only because some cycles moved away from the price peak at 09:00. Its explanation nevertheless states that the schedule "was optimized to run appliances during lower price slots", a claim that nothing checks. The pipeline's run took 19.2 seconds,

15 909 tokens and $0.003 in API fees, with 18 tool calls, and its schedule passed the verification step without a feedback round.

### 5.3.1 Design iterations

The final pipeline is the result of eight build-and-evaluate cycles, in line with the iterative design and demonstration activities of DSR (Peffers et al., 2007). The first five, summarised here, built the base pipeline of the earlier design. The first four used a pilot of six scenarios of the main benchmark, one per type (A01, B03, C02, D01, E01 and F04); the fifth used the full main benchmark. Because the pipeline was refined on these scenarios, the main benchmark is treated as exploratory, and the hypotheses of that design were confirmed on the held-out Q set (Appendix B). The last three iterations, which added the cost check, retrieval, the optimiser tool and the completion step, are reported in Section 5.5, and a ninth, made after the registered test to shorten the response time, in Section 5.7. Table 5.5 summarizes the first five. Every change is a general mechanism that applies to all scenarios and, where it concerns errors of the language model, equally to the single agent; none was tuned to an individual scenario.

Three observations follow from the iterations. First, most problems concerned reliability rather than cost: the agents usually found cheap start times, but a schedule had to be complete, within the peak limit and within every window before its cost counted. This is why the verification step grew from a peak-load check into a check of all three conditions. Second, the changes that helped GPT-4o-mini did little for the small local model. In every complete pilot from iteration 2 onwards its pipeline produced only one or two valid schedules out of six, while the single agent with the same model reached six out of six once malformed outputs were recovered. Third, the verification step can trade cost for validity. Only seven of the 144 runs of iteration 5 needed a feedback round, and in one of them (C08, run 3) the repaired schedule was valid but 17.3 % above the optimum, against 0.4 and 0.5 % in the two runs that needed no repair. Over the whole benchmark the window check improved validity and the mean gap, but the change in gap is not statistically significant (Wilcoxon $p = 0.65$, $N = 48$).

**Table 5.5 Design iterations of the pipeline (pilot: six scenarios, one run each; iteration 5: 48 scenarios, three runs each).**

| | Problem observed | Change | Effect |
|---|---|---|---|
| 1 | The Resolver moved appliances back and forth until it ran out of steps, and the final summary could report success for a schedule that still exceeded the peak limit. | Automatic mode of `adjust_schedule`; the summary is written from the result of a final peak-load check. | Pipeline with GPT-4o-mini: valid schedules 3/6 to 6/6, mean gap 11.4 % to 0.6 %. |
| 2 | With the open-weight models, the provider rejected malformed tool calls, and the local model often ended its turn in prose without the required structured output. | Recovery nudge (at most two per agent and run) for both cases, applied identically to every agent and logged. | Single agent with gpt-oss-20b: valid schedules 4/6 to 6/6 once prose endings were also covered. |
| 3 | The Orchestrator's task brief sometimes omitted an appliance, which was then never scheduled. | The daily plan travels with the brief to the Profiler; the verification step checks that every appliance has a start and returns missing ones to the Scheduler. | Pipeline with GPT-4o-mini: mean gap 0.63 % to 0.36 %, with feedback rounds in some runs; no improvement with gpt-oss-20b. |
| 4 | With the step budget divided equally (25 steps per specialist), the Scheduler ran out of steps. | Budget divided by workload: 15 for the Profiler, 35 for the Scheduler and 25 for the Resolver, still 75 in total. | Scheduler step-limit failures 3 to 0 across the pilot; local pipeline valid schedules 1/6 to 2/6. |
| 5 | On the full benchmark, the only two invalid pipeline schedules started a tumble dryer cycle outside its time window. | The verification step also checks every start against its window and returns violations to the Scheduler. | Valid schedules 142/144 to 144/144, mean gap 1.50 % to 1.25 %, type E 2.30 % to 0.21 %. |

## 5.4 Evaluation of the earlier design

Before the redesign described in Section 3.3, the study compared the base pipeline with a single ReAct agent and measured each schedule by its gap to the optimal cost. Appendix B reports that evaluation in full: the exploratory tests on the first benchmark, the comparison with the baselines and real tools, and the first confirmatory run on the held-out Q set. Three of its results shaped the final design.

First, tools made the agents viable. With GPT-4o-mini, both agent designs came within 1.5 % of the optimal cost on average, whereas the same model answering without tools was 18.8 to 23.1 % above it and captured at most a quarter of the achievable saving. Second,

splitting the task among specialist agents did not improve the schedules. The pipeline was equivalent to the single agent within one percentage point of the optimum, used about 30 % fewer input tokens and cost 21 % less per run, but it was 52 to 69 % slower. Third, the agents searched less well than simple methods on the hardest days. The smart timer came significantly closer to the optimum than either agent design, and both designs lost most on the days on which five appliances shared one window under the peak limit, with mean gaps of 3.9 to 6.5 %. These results led to the redesign: the single agent was dropped, the saving against no scheduling became the outcome, and the pipeline was given retrieval of solved days and the optimiser tool.

## 5.5 Exploratory results on the factorial benchmark

After the redesign, every system was run on the exploratory main set of 96 scenarios (Section 5.1.4), the systems with a language model three times per scenario. The run served two purposes: it showed what each design step contributed, and it checked the hypotheses before they were registered. Its results are exploratory. One iteration before it was dropped. Iteration 6 sent a valid schedule back to the Scheduler once when it cost more than 3 % above a lower bound that ignores the peak limit. On the first benchmark it lowered the mean gap on type C from 0.84 to 0.12 %, but raised it on type D from 5.9 to 9.1 % and overall from 1.25 to 1.65 %, with 12 % more tokens, so it was not kept. Table 5.6 follows the pipeline through its last design iterations. The base pipeline was valid in every run and saved 26.72 % of the cost without scheduling, against 27.59 % for the optimum. Retrieval of solved days closed half of the remaining gap but doubled the tokens per run. The optimiser tool brought the saving to within 0.03 percentage points of the optimum and the tokens back to the level of the base pipeline, but in 2 of 288 runs the final schedule left out one appliance after the feedback round. Iteration 8 therefore added a deterministic completion step to the verification (Section 4.4); in the rerun of all 288 runs it was not needed once, every schedule was valid, and the saving equalled the optimum to two decimals.

**Table 5.6 Last design iterations of the pipeline on the exploratory factorial benchmark (GPT-4o-mini, 96 scenarios, three runs each; saving pooled over all runs).**

| Design iteration | Valid | Saving | Gap to optimum | Tokens per run | Mean latency |
|---|---|---|---|---|---|
| Base pipeline | 100 % | 26.72 % | 0.76 % | 14 891 | 26.0 s |
| + retrieval of solved days | 100 % | 27.09 % | 0.36 % | 26 487 | 21.8 s |
| + optimiser tool (iteration 7) | 99.3 % | 27.56 % | 0.12 % | 14 134 | 16.9 s |
| + completion step (iteration 8, final) | 100 % | 27.58 % | 0.01 % | 13 819 | 21.5 s |
| Optimum (upper bound) | 100 % | 27.59 % | 0 | – | – |

The other systems on the same benchmark show the pattern that the registered hypotheses later tested. EMHASS with measured power profiles saved 27.36 % and the smart timer

27.50 %, both with valid schedules in every scenario. The cheapest-hours automation, its hourly variant and the cheapest window saved 12.4 to 13.8 % and were valid in 68 to 71 % of the scenarios; on heavy-load days under the strict limit none of their schedules was valid. The LLM without tools saved 7.8 % and was valid in 92 % of its runs. Applied to these data, all seven tests of the second pre-registration were supported.

## 5.6 Confirmatory evaluation on the held-out set Q2

The hypotheses of Table 3.2 were fixed in the second pre-registration, which was committed to the project repository on 6 October 2026 together with the code, prompts, tools, case library and analysis script. The run on Q2 followed the same day. The Multi-Agent Pipeline and the LLM without tools used GPT-4o-mini with three runs per scenario, EMHASS and the rule-based systems one run each, and the analysis was run exactly as registered. The run cost about USD 0.84 in API fees.

Two deviations from the run protocol occurred. First, the campaign was by mistake started a second time while the first process was still running, and for a few minutes both processes worked on the same plan. Two runs, the third run of scenario Q2-VH04-80 for the pipeline and for the LLM without tools, were therefore recorded twice. The analysis keeps the first record of every run, as it does in all evaluations, so each run enters the tests once; both records remain in the results file. Second, from about 15:30 on the day of the run, the response time of both language model systems rose by a factor of about 4.5 at the same moment: the median latency of the pipeline went from 19.4 to 94.3 seconds and that of the LLM without tools from 2.6 to 11.6 seconds, while the tokens per run (14 263 before and 14 056 after) and the tool calls per run (15.6 and 15.2) of the pipeline did not change. The slowdown therefore lay outside the system, in the response time of the language model service or the network. The run was not repeated, since the registered test applies to the run as it happened. Table 5.7 reports the seven registered tests.

Five of the seven hypotheses are supported. For SQ1, the pipeline saved on average 15.4 percentage points more of the cost without scheduling than the same language model answering without tools (H1a), with a large effect. Its API cost was \$0.0025 per recommendation, with an upper 95 % confidence bound of \$0.0026, far below the limit of \$0.10 (H1c). Two hypotheses are not supported. The pipeline was valid in all 288 runs, but the LLM without tools was valid in 96.5 % of its runs, so the difference of 3.5 percentage points in the validity rate is not significant after the Holm adjustment (H1b; unadjusted $p = 0.046$). The pre-registration had named this test as the one with the least power. The mean latency of 63.5 seconds, with an upper bound of 66.5 seconds, did not meet the objective of 60 seconds (H1c). Before the slowdown described above, the median latency was 19.4 seconds, in line with the 21.5 seconds of the exploratory run, so the result reflects the response time of the language model service on the day of the run as much as the design. It is nevertheless the registered result.

For SQ2, the pipeline saved 8.6 percentage points more than the cheapest window (H2a) and 10.0 more than the cheapest-hours automation (H2b), both with large effects, and

**Table 5.7 Pre-registered confirmatory tests on the held-out set Q2 (GPT-4o-mini, $N = 96$ scenarios; estimate = mean paired difference in percentage points of the cost without scheduling, or the mean for a bound; $p$-values Holm-adjusted within each sub-question).**

| | Hypothesis | Estimate | $p$ | $r$ | Supported |
|---|---|---|---|---|---|
| *SQ1, value and cost* | | | | | |
| H1a | Pipeline saves more than the LLM without tools | $+15.4$ pp | $< 0.001$ | 0.87 | Yes |
| H1b | Pipeline is valid more often than the LLM without tools | $+3.5$ pp | 0.091 | 0.20 | No |
| H1c | Mean latency per recommendation below 60 s | 63.5 s | 0.97 | – | No |
| H1c | Mean API cost per recommendation below \$0.10 | \$0.0025 | $< 0.001$ | – | Yes |
| *SQ2, conditions* | | | | | |
| H2a | Pipeline saves more than the cheapest window | $+8.6$ pp | $< 0.001$ | 0.85 | Yes |
| H2b | Pipeline saves more than the cheapest-hours automation | $+10.0$ pp | $< 0.001$ | 0.86 | Yes |
| H2c | Pipeline saves the same as EMHASS ($\pm 1$ pp) | $+0.10$ pp | $< 0.001$ | – | Yes |

it saved the same as EMHASS within the margin of $\pm 1$ percentage point (H2c). The mean difference to EMHASS was $+0.10$ percentage points with a 90 % confidence interval of $[0.04, 0.16]$, which also lies inside the narrower margin of $\pm 0.5$ points; the Wilcoxon version of the test agrees at all three margins. A two-sided Wilcoxon test detects the small difference in favour of the pipeline ($p = 0.001$), which reflects the default solver tolerance of the EMHASS service and is far too small to matter for a household. Table 5.8 summarises all systems.

**Table 5.8 All systems on the held-out set Q2 (96 scenarios; saving pooled over all runs; validity = share of valid runs; latency and API cost per run).**

| System | Valid | Saving | Gap to optimum | Mean latency | API cost per run |
|---|---|---|---|---|---|
| Multi-Agent Pipeline | 100 % | 27.40 % | 0.00 % | 63.5 s | \$0.0025 |
| Smart timer (own heuristic) | 100 % | 27.34 % | 0.05 % | $< 0.01$ s | – |
| EMHASS, measured profiles | 100 % | 27.30 % | 0.12 % | 0.9 s | – |
| EMHASS, nominal power | 100 % | 25.39 % | 2.17 % | 0.9 s | – |
| Cheapest window | 65.6 % | 12.49 % | 15.37 % | $< 0.01$ s | – |
| Cheapest-hours automation | 68.8 % | 10.71 % | 17.90 % | $< 0.01$ s | – |
| Cheapest hours, hourly prices | 66.7 % | 10.09 % | 17.99 % | $< 0.01$ s | – |
| LLM without tools | 96.5 % | 6.24 % | 25.79 % | 6.8 s | \$0.0004 |
| Optimum (upper bound) | 100 % | 27.40 % | 0 | – | – |

The second part of SQ2 asks under which conditions the pipeline saves more than, or as much as, the existing tools. Table 5.9 breaks the saving down by each factor; these results are descriptive. The pipeline reached the saving of the optimum under every price pattern, load and limit, and EMHASS was within 0.3 percentage points of it throughout. The gap to the cheapest-hours tools depends mainly on the limit and the load. Under the realistic limit of 8 kW they saved 18 to 22 %, against 27.9 % for the pipeline, because their schedules were mostly valid and only less well placed. Under the strict limit of 3.5 kW they saved 2.4 to 3.0 %, because their schedules often exceeded the limit and were scored at the cost without scheduling. Of the 30 scenarios that require coordination, only 3 to 13 % of their schedules were valid, against 100 % for the pipeline, EMHASS and the smart timer and 97 % for the LLM without tools. Paired by day, moving from the strict to the realistic limit raised the mean saving of the cheapest-hours automation and the cheapest window by 7.9 and 9.7 percentage points, but that of the pipeline by only 0.4. Across price patterns, the differences were largest on volatile and spike days, where the pipeline saved 37.0 and 34.2 % against 10.9 to 17.0 % for the cheapest-hours tools, and smallest on days with negative prices, on which little could be saved by any system.

**Table 5.9 Saving on the held-out set Q2 by condition (percent of the cost without scheduling, pooled over all runs of the scenarios in each group). Each factor is averaged over the other two.**

| Condition | Optimum | Pipeline | EMHASS | Cheapest hours | Cheapest window | LLM, no tools |
|---|---|---|---|---|---|---|
| Stable prices | 9.0 | 9.0 | 8.8 | 3.0 | 4.1 | 4.2 |
| Volatile prices | 37.0 | 37.0 | 36.9 | 12.1 | 14.3 | 6.5 |
| Price spike | 34.2 | 34.2 | 34.2 | 14.9 | 17.0 | 8.4 |
| Negative prices | 4.1 | 4.1 | 3.8 | 2.3 | 3.0 | 0.5 |
| Light load | 23.2 | 23.2 | 23.2 | 13.5 | 16.4 | 1.7 |
| Heavy load | 28.6 | 28.6 | 28.4 | 9.9 | 11.4 | 7.5 |
| Realistic limit (8 kW) | 27.9 | 27.9 | 27.8 | 18.4 | 22.2 | 6.6 |
| Strict limit (3.5 kW) | 26.9 | 26.9 | 26.8 | 3.0 | 2.8 | 5.9 |

Two systems reported without hypotheses complete the picture. The smart timer, the rule written for this study, reached 27.34 %, slightly below the pipeline (mean difference 0.03 percentage points; unregistered Wilcoxon test, $p = 0.034$), with a valid schedule in every scenario and no language model. EMHASS configured with a constant nominal power saved 25.39 %, 1.4 percentage points less than the pipeline ($p < 0.001$), because a constant power at the cycle's peak overstates the load of the cycle and rules out start times at which the real cycle would fit under the limit. Summed over the 96 scenarios, with each scenario counted once as the mean of its runs, the pipeline saved € 19.15, EMHASS € 19.08, the cheapest-window rule € 8.73, the cheapest-hours automation € 7.48 and the LLM without tools € 4.36.

In summary, on days it had not seen and with fifteen-minute prices, the Multi-Agent Pipeline produced a valid schedule every time and saved as much as the optimum and as EMHASS, far more than the cheapest-hours tools that plan each appliance on its own. Its

advantage over these tools comes from coordinating the appliances under the household's peak limit and is largest where that limit binds. It did not beat the LLM without tools on validity by a significant margin, and it missed the latency objective on the day of the run. Appendix A repeats the comparison with UK prices and a UK tariff of the period in which the REFIT data were recorded; the savings are smaller, but the ranking of the systems is the same.

## 5.7 Post-registration iteration: response time

The pipeline missed the registered objective for its response time (H1c). To find where the time goes, three of its runs on Q2 were profiled after the confirmatory run. About 97 % of each run was spent waiting for eleven to thirteen language model calls made one after the other, each taking one to three seconds. The tools, including the optimiser, took less than half a second together. The Profiler alone took six to ten seconds, although its work, reading the household data and classifying each appliance, is deterministic.

Design iteration 9 therefore reduces the number of sequential calls without changing the model, the tools or the optimiser. The Profiler receives the household data and the classification of every appliance in its message, as produced by the same tools, and answers in one call. The Scheduler starts with the optimiser tool and returns its start times without checking each cost again, since the verification step checks the schedule anyway. The Resolver is called only when the Scheduler's schedule fails the verification step. Finally, the Orchestrator's summary is given the cost of the schedule and its saving, which the summaries of the final pipeline did not state (Section 5.3). Because the change was made after the registration, it is evaluated as an exploratory comparison and recorded as a note in the pre-registration; the registered result of H1c stands. The lean variant and the final pipeline were each run three times on the 96 scenarios of the exploratory main set, alternating scenario by scenario so that both met the same conditions of the language model service. Table 5.10 reports the result.

The lean variant halved the response time. Its mean latency was 9.6 seconds against 18.0 seconds, it was faster in 282 of the 288 paired runs (Wilcoxon $p < 0.001$), and its slowest run took 25 seconds. It used 62 % fewer tokens and cost 59 % less per recommendation. The schedules did not get worse: every run of both variants was valid, the two returned schedules of the same cost in 285 of the 288 paired runs, and in the other three the lean variant was cheaper, in two of them by 1.4 percentage points where the final pipeline had missed the optimum. Every summary of the lean variant stated the cost, and 285 of them also the saving. If a slowdown of the language model service lengthens every call by the same factor, as it did during the confirmatory run, the factor of 4.5 observed there would take the lean variant to about 43 seconds and the final pipeline to about 81 seconds. The lean variant would then still meet the objective of 60 seconds, although this was not tested on the held-out set.

**Table 5.10 Final pipeline (iteration 8) and lean variant (iteration 9), interleaved on the exploratory main set (GPT-4o-mini, 96 scenarios, three runs each).**

| Measure | Final pipeline (iteration 8) | Lean variant (iteration 9) |
|---|---|---|
| Valid runs | 288/288 | 288/288 |
| Saving (pooled) | 27.58 % | 27.59 % |
| Gap to optimum | 0.01 % | 0.00 % |
| Mean latency | 18.0 s | 9.6 s |
| Median latency | 18.0 s | 9.2 s |
| 95th percentile of latency | 25.8 s | 15.1 s |
| Tokens per run | 14 101 | 5 405 |
| API cost per run | $0.0025 | $0.0011 |
| Tool calls by the agents per run | 15.4 | 4.1 |
| Summaries stating the cost | 0/288 | 288/288 |

# 6. Discussion

This chapter interprets the results of Chapter 5. It first answers the two sub-questions and the main research question, then relates the findings to the prior work reviewed in Chapter 2 and draws implications for households and for the design of agent-based home energy management. It ends with the limitations of the study and directions for future research.

## 6.1 Interpretation of the research questions

### 6.1.1 SQ1: value and cost of the recommendations

SQ1 asked how much the pipeline's recommendations save, how often they are valid, and what they cost to produce, compared with a language model that answers without tools. On the held-out days, the pipeline saved 27.4 % of the cost of starting every appliance when ready, the same as the optimal schedule, while the same model answering in a single prompt saved 6.2 %; the mean paired difference was 15.4 percentage points (H1a). The difference does not come from the language model, which is the same in both systems, but from how it is used. Without tools the model has to read 96 prices, add up power profiles and compare start times in its head, and it rarely finds a cheap combination; with tools those calculations are exact and the model only decides what to ask for. This supports the premise of the ReAct approach, that reasoning grounded in tool observations does better than unaided generation (Yao et al., 2023), on a task where the observations are numbers.

Validity told a different story than expected. The LLM without tools produced a valid schedule in 96.5 % of its runs, so most of its failures were expensive schedules rather than schedules that broke a constraint, and the pipeline's 100 % validity was not significantly higher (H1b). The prompt gave the model every window and the limit explicitly, and on most days a schedule that starts each appliance near the beginning of its window is valid. Validity is therefore a weak differentiator on this benchmark. The pipeline's value for validity lies less in the average than in the check: its verification step tests every schedule before it is returned and sends a flawed one back, and a missing appliance is placed deterministically after the last round. None of the 576 runs of the final design returned an invalid schedule.

The cost of a recommendation was about a quarter of a cent in API fees, forty times below the objective of $0.10 (H1c). On the held-out days the average saving was about €0.20 per household and day, so the inference cost was around one percent of the saving. The response time missed its objective: the mean was 63.5 seconds against a limit of 60. The run shows that this was caused by the language model service, which slowed down by a factor of about 4.5 for both language model systems during the run while the work done per run stayed the same. Before the slowdown the median was 19 seconds. For

a day-ahead recommendation, which is computed once after the prices are published in the early afternoon, a minute is not a practical problem, but the result shows that the speed of the language model service is not under the designer's control. The number of calls made to it is. Profiling after the run showed that almost all of the time was spent waiting for eleven to thirteen model calls made one after the other, and a later design iteration that left out the calls with nothing to decide halved the response time, to 9.6 seconds on the exploratory set, while using fewer tokens and returning the same schedules (Section 5.7). The registered result stands, but the response time is largely a consequence of the design rather than a fixed cost of using agents.

### 6.1.2 SQ2: conditions under which the system pays off

SQ2 asked under which price patterns, household loads and peak limits the recommendations save more than, or as much as, the existing home energy management tools available to households. The comparison with the cheapest-hours automations depends mainly on the peak limit and the household load. These tools plan each appliance on its own. On light-load days they saved 13.5 to 16.4 % against 23.2 % for the pipeline, and under the realistic limit 18.4 to 22.2 % against 27.9 %, because their schedules were mostly valid and only less well placed. Where several appliances had to be staggered under a tight limit, they almost always exceeded it: of the 30 held-out scenarios that required coordination, only 3 to 13 % of their schedules were valid. Under the strict limit they therefore saved about 3 % against 27 %. The price pattern changes the size of the stakes rather than the ranking: on volatile and spike days the pipeline saved 34 to 37 %, on stable and negative-price days 4 to 9 %. The answer to the first half of SQ2 is therefore that the pipeline saves more than the cheapest-hours tools under every condition, and much more where the household runs several appliances under a binding limit.

Against EMHASS, the pipeline saved as much under every condition: the mean difference was 0.1 percentage points (H2c), within 0.3 points in every group of Table 5.9. This is expected, because both solve the same optimization problem in the end, EMHASS with its own solver and the pipeline through its optimiser tool. The difference between them lies outside the cost. EMHASS has to be configured with the power, run time and window of each appliance in numeric form, and it returns a power plan per time step without explanation. The pipeline works from the day's appliance plan, checks its own schedule against the limit and the windows, and returns start times with a short explanation for the household. Whether households value that explanation was not measured in this study.

A rule written for this study, the smart timer, placed the appliances one at a time and came within 0.1 percentage points of the pipeline without a language model. On problems that are fully specified, with an explicit window for every appliance, a well-chosen greedy rule is thus almost as good as an optimizer. This qualifies the result for SQ2: the advantage of the pipeline over the cheapest-hours tools is an advantage of planning the appliances together under the limit, which a simple peak-aware rule also has, and not of the language model as such.

### 6.1.3 RQ: design of the multi-agent system

The main research question asked how a hierarchical ReAct multi-agent system can be designed to give households reliable, cost-saving daily recommendations. The design iterations answer it as much as the final results do. The first designs left the search for a good schedule to the agents, with tools that move one appliance at a time. They found the optimum on most days but not on the hardest ones: on one winter day of the first benchmark the pipeline ended 30 % above the optimal cost with a schedule that no single move could improve. Telling the agents that a schedule was too expensive helped on moderately difficult days but not on that one, because the problem was the search, not the agents' awareness of the cost. Retrieval of similar solved days halved the remaining gap but doubled the tokens per run. Only the optimiser tool closed the gap, and it also cut the response time and the tokens back to the level of the base pipeline. The last iteration added a deterministic completion step after the model occasionally left an appliance out, and an iteration after the registered test removed the model calls that had nothing to decide, which halved the response time.

Five design principles follow from these iterations. First, every number the agents reason with should come from a deterministic tool, and the combinatorial search should be handed to a solver; the language model is good at deciding what to compute and at explaining the result, not at searching a space of start-time combinations. Second, every schedule should be checked by a verification step that does not rely on the agents, and returned for revision when it fails. Third, rare but predictable failures of the model, such as a missing appliance, are better closed by a deterministic fallback than by more prompting. Fourth, the roles of the agents should follow the structure of the problem: which appliances can move, when each is cheapest, and how they fit together under the limit. Fifth, a model call should be made only where the model has something to decide. Deterministic reads belong in the agent's message rather than in separate tool turns, and a specialist whose input already passes the verification step need not be called, because the response time grows with every call made one after the other. Under these principles the multi-agent system reached the optimal saving with a valid schedule in every held-out run, at a fraction of a cent per recommendation.

## 6.2 Relation to prior work

The results address the open problems that Jonah et al. (2026) listed for LLM-based home energy management. Multi-appliance coordination, which failed for two of the three models of El Makroum et al. (2025), did not fail once in the final design, but only because the coordination was delegated to a solver; the earlier designs, in which the agents coordinated the appliances themselves, produced valid schedules but fell short of the optimum on the days that required coordination, by up to 30 %. Jonah et al. recommend that an independent deterministic validator check every schedule before it is executed; this thesis implements that recommendation as the verification step. It detected the two exploratory runs in which the model left an appliance out, and the completion step added in

the last design iteration closes that gap without relying on the model. Their agents captured 96.7 to 98.0 % of the savings available between an off-peak timer and an optimum; the final pipeline here captured all of the saving available between no scheduling and the optimum, on a held-out benchmark that varies the price pattern, the load and the limit rather than a single household over a few weeks.

The final design follows the division of work of OptiGuide (Li et al., 2023), in which a language model uses an optimization solver instead of replacing it. The thesis adds evidence that this division also holds for a hierarchy of agents and for a household-scale problem, and that it improves cost, tokens and response time at once compared with agents that search themselves. Retrieval-augmented generation (Lewis et al., 2020) helped less: solved examples improved the agents' schedules on moderately difficult days but did not enable them to search hard ones, which suggests that examples teach the form of a good solution more easily than the search for it. For multi-agent systems in general (Guo et al., 2024; Hong et al., 2024), the results support the use of specialised roles with structured hand-offs, but they also show that the roles add no cost advantage on their own; the gain came from what the roles were given to work with.

## 6.3 Implications for practice

For households, the results suggest caution with the cheapest-hours automations that are widely shared for Home Assistant and Shelly relays. They work when the household runs one or two appliances and has power to spare, but they ignore the household's total load and exceeded a tight limit on most days that required coordination. Under the power-based distribution tariffs that Finnish network companies are moving towards, which charge for the household's highest hourly power above a threshold, such tools would raise the power charge exactly on the days when several appliances run. A peak-aware method, whether an optimizer such as EMHASS, a peak-aware rule or an agent with an optimiser tool, avoids that.

For developers of home energy management systems, the results argue for a hybrid design rather than for replacing optimization with language models. The solver guarantees the best schedule; the language model adds the interface: it can take the day's plan, check and explain the result, and could in future take wishes stated in ordinary language. The cost of the language model is small compared with the savings. Its response time depends on a cloud service, so the recommendation should be computed ahead of time rather than on request, and on the number of model calls made one after the other, which the design should keep small.

## 6.4 Limitations

The study has seven main limitations. First, the evaluation is a simulation on historical data. Real consumption profiles and real Nord Pool FI prices make it more realistic than

synthetic scenarios, but the system was not tested in a live deployment, where appliance states change during the day and users may not keep to the windows they gave.

Second, the appliance data come from UK homes. The REFIT dataset was collected from 20 UK houses between 2013 and 2015, of which 18 are used, and may not reflect the appliances and habits of Finnish households today. No public Finnish dataset measures washing machines, dishwashers and tumble dryers separately over time. Fingrid's Datahub open data publish the hourly consumption of Finnish metering points only in aggregated form, by user group (Fingrid, 2024), and the Finnish household electricity use study reports measured annual consumption per appliance group from about one hundred homes, without load time series (Adato Energia, 2008). Appliance-level datasets from other countries, such as a Norwegian dataset of a single household over 24 days, are too small to replace REFIT. The cycle profiles should transfer, because the heating phase that dominates the energy and peak power of a wash cycle depends on the machine and not on the country. The base load transfers less well. Many Finnish homes have electric space heating, an electric sauna or a heat pump, whose power far exceeds that of the UK homes in REFIT, so in a real Finnish house the peak limit, in particular the realistic 8 kW limit, would bind more often than in these scenarios. Pricing the REFIT cycles at UK prices and a UK tariff of their own period instead gives smaller savings but the same ranking of the systems (Appendix A).

Third, LLM output is non-deterministic, and two runs of the same scenario may produce different schedules. Each scenario is run three times and the results are averaged, which reduces but does not remove this variance. Fourth, only three appliance types are studied; electric vehicle chargers, heat pumps and battery storage would need their own evaluation. Fifth, the daily appliance plan is given in structured form, with an explicit window for every appliance. The study therefore does not test whether the agents can interpret wishes stated informally, which is where they are expected to have an advantage over rule-based scheduling. On these fully specified problems, the smart timer, a simple rule written for this study, came within 0.1 percentage points of the pipeline on the held-out set (Section 5.6), and on the first benchmark it saved more than the earlier agent designs (Appendix B).

Sixth, the final design was evaluated with one language model, GPT-4o-mini, through one cloud service, and the held-out run took place on a single day. Its response time therefore reflects that service on that day, as the slowdown during the run showed, and the results may differ with other models. The faster variant of design iteration 9 was evaluated on the exploratory set only, not on held-out days. A small model run locally was explored only for an earlier design. Seventh, the peak limit was a hard constraint in every benchmark, whereas real power-based tariffs price the peak instead of forbidding it. Only the demonstration application lets a schedule exceed the limit, charging a fee for each kilowatt above it; none of the reported results use that fee. The explanations that the pipeline gives were not evaluated with users. The study shows that the pipeline produces an explanation for every schedule, not that households understand or trust it more than the plan of an optimizer.

## 6.5 Future research

Several directions for future research emerge from this study. First, extending the pipeline to thermal loads such as electric space heating, heat pumps and domestic hot water systems would address the gap explicitly identified by El Makroum et al. (2025). Second, implementing the price API and data reader as MCP servers would enable reuse by future researchers without custom integration code, contributing to the emerging standardization of agent tool interfaces. Third, a live deployment study involving Finnish households on spot-price contracts would provide ecological validity that simulation-based evaluation cannot. Fourth, evaluating the final design with other language models, including small open-weight models run locally, would show whether its results depend on GPT-4o-mini; since the optimiser tool takes over the search, a small model may be sufficient for the remaining work of formulating, checking and explaining. Fifth, a benchmark of informal household requests with annotated constraints would test the agents where a rule-based scheduler cannot operate, by measuring how accurately each system turns such requests into a correct schedule. Sixth, the hard peak cap of the benchmark could be replaced by the power charge of a real distribution tariff, so that a system weighs the energy saving of a cheap hour against the cost of raising the household's monthly peak, as under the Energy Authority's guidance for power-based distribution pricing. The demonstration application of this thesis already prices the peak in this way, with an optimum that includes the fee, which could serve as a starting point. Seventh, a user study should test the explanations: whether households understand the pipeline's recommendations, trust them more than a numeric plan, and follow them, which is the advantage over an optimizer that this thesis argues for but did not measure.

# 7. Conclusions

Households on dynamic electricity contracts can lower their bills by running flexible appliances in the day's low-price hours, but planning several appliances against a changing price curve and a limit on the household's total power is hard to do every day without help. This thesis designed and evaluated, following Design Science Research, a hierarchical ReAct multi-agent system that gives a household a daily appliance recommendation. Four agents built in LangGraph, an Orchestrator, an Appliance Profiler, a Price-Aware Scheduler and a Conflict Resolver, work from the day's appliance plan and prices with deterministic tools, are shown similar past days whose best schedules are known, and hand the search for the cheapest schedule to an optimiser tool. A verification step checks every schedule before it is returned. The system was evaluated on appliance cycles measured in the REFIT homes and Finnish day-ahead prices, first on an exploratory benchmark and then on a held-out benchmark of 96 scenarios with hypotheses registered in advance.

## 7.1 Answers to the research questions

**SQ1** asked how much the recommendations save, how often they are valid, and what they cost to produce, compared with a language model answering without tools. On the held-out days the pipeline saved 27.4 % of the cost of starting every appliance when ready, the same as the optimal schedule, against 6.2 % for the same model without tools. All 288 of its schedules were valid, but so were 96.5 % of those of the model without tools, and the difference in validity was not significant. A recommendation cost about a quarter of a cent in API fees. The mean response time of 63.5 seconds missed the objective of one minute because the language model service slowed down during the run; before the slowdown the median was 19 seconds. A design iteration after the test, which leaves out model calls with nothing to decide, halved the response time to 9.6 seconds on the exploratory set without changing the schedules. Tool use, not the language model alone, is what makes the recommendations valuable.

**SQ2** asked under which price patterns, household loads and peak limits the recommendations save more than, or as much as, existing home energy management tools. The pipeline saved 8.6 to 10.0 percentage points more than the cheapest-hours automations shared for Home Assistant and Shelly relays, and the gap was largest where several appliances had to share a tight limit: there the automations, which plan each appliance on its own, rarely produced a schedule within the limit. On light-load days with power to spare the difference was smaller. The pipeline saved the same as the optimizer EMHASS under every price pattern, load and limit, within 0.1 percentage points on average. A simple peak-aware rule written for this study came almost as close, which shows that the advantage over the cheapest-hours tools comes from planning the

appliances together under the limit.

The main **RQ** asked how such a system can be designed to give reliable, cost-saving daily recommendations. The design iterations show that agents which search for a schedule themselves fall short on the days that need coordination, and that solved examples help only partly. The system became reliable and optimal once the arithmetic was done by deterministic tools, the combinatorial search by a solver, and every schedule was checked independently of the agents, with a deterministic fallback for the rare case in which the model leaves an appliance out. The language model's part is to formulate the problem, check the result and explain it to the household.

## 7.2 Contributions

The thesis makes three contributions. The first is the artifact: the Multi-Agent Pipeline, together with the documented design iterations that led to it and five design principles drawn from them (Section 6.1). The second is an evaluation of what its recommendations are worth and cost, against a language model without tools, tested with pre-registered hypotheses on held-out days. The third is a map of the conditions under which an agent-based recommendation saves more than, or as much as, the home energy management tools households can already install, built on a factorial benchmark that varies the price pattern, the household load and the peak limit separately. The benchmarks, the case library, the code, the pre-registrations and all run records are published in the project repository, so the results can be reproduced and extended.

## 7.3 Implications

For research on language model agents, the results add to the evidence that agents are most reliable when they use solvers and checks rather than replace them, and they show this for a hierarchy of agents and a household-scale scheduling problem. They also show that comparing an agent only with a language model without tools, or only with an optimum, hides the most relevant comparison: the simple tools and rules that users already have. For practice in Finland, where a third of households had a dynamic-price contract at the end of 2024 (Energy Authority, 2025) and network companies are moving towards power-based distribution tariffs, the results suggest that the cheapest-hours automations popular today will become costly on days when several appliances run, and that any recommendation should account for the household's total power. An agent-based system can match an optimizer on cost at a negligible API cost per day, and its value over an optimizer lies in the explanation and in accepting the household's plan without numeric configuration.

## 7.4 Limitations and further work

The evaluation is a simulation based on UK appliance data and one language model, with the peak limit as a hard constraint, and the explanations were not tested with users (Section 6.4). The most important next steps are a field study with Finnish households, a test of whether households understand and act on the explanations, a benchmark of wishes stated in ordinary language, where a rule-based tool cannot follow, and a version of the problem in which the peak is priced as in a real power-based tariff (Section 6.5).

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# A. Robustness check: UK prices of the REFIT period

The benchmarks of this thesis combine appliance cycles recorded in UK homes in 2013–2015 with Finnish prices of 2022–2026 (Section 5.1). This appendix checks whether the results also hold when the appliance data are priced in their own country and period. It is a robustness check that was not pre-registered, and its results are exploratory.

## A.1 Prices, tariff and scenarios

The ENTSO-E Transparency Platform publishes day-ahead prices for the Great Britain bidding zone only from 17 December 2014, in GBP per MWh (ENTSO-E, 2026). This covers the last months of the REFIT recordings, which ended in mid-2015 (Murray et al., 2017). The prices of 18 December 2014 to 31 July 2015, hourly, give 224 complete days on London time. They were converted to euros at the mean of the European Central Bank's daily reference rates for the same period, 0.73161 GBP per EUR (European Central Bank, 2026), so that the agents' tools and the costs use the same currency as in the other benchmarks. The saving in percent does not depend on the currency.

The consumer price follows the form of the Finnish tariff (Section 5.1): the spot price plus a constant charge per kilowatt-hour, with VAT. The constant charge stands for all non-wholesale costs of a UK household in 2015, that is network costs, policy costs, the supplier's costs and margin, and the standing charge spread over consumption. It was set so that the average consumer price equals the average UK unit cost of standard electricity in 2015, 15.4 pence per kWh including 5 % VAT, which follows from DECC's average bill of £584 for an annual consumption of 3 800 kWh (Department of Energy & Climate Change, 2016). With a mean spot price of 4.05 pence per kWh in the window, the charge is 10.62 pence per kWh, or 0.1452 euros per kWh. In 2014–2015 almost no UK household had a dynamic hourly tariff, so the set describes a hypothetical household on such a tariff rather than a real one.

The scenarios follow the factorial design of Section 5.1.4 with one change. The window contains no day with a negative price, so the set has three price patterns, stable, volatile and spike, each defined relative to the days of the window. With two household loads, two peak limits and four days per pattern and load, the set has 48 scenarios, 14 of which require coordination. The cost of starting every appliance when it is ready has a median of € 0.63 per scenario. Every system of the confirmatory evaluation was run on the set in the same way as on Q2: the Multi-Agent Pipeline and the LLM without tools three times per scenario with GPT-4o-mini, and EMHASS and the rule-based systems once.

## A.2 Results

Table A.1 summarises the results. The ranking of the systems is the same as on the Finnish benchmarks. The pipeline produced a valid schedule in all 144 runs and saved 6.69 % of the cost without scheduling, the same as the optimal schedule. EMHASS saved 6.45 % and the smart timer 6.55 %. The cheapest-hours automations saved 2.6 to 3.6 %, and on the 14 scenarios that require coordination only 7 % of their schedules were valid, against 100 % for the pipeline, EMHASS and the smart timer. The LLM without tools saved 0.18 %, and a quarter of its runs cost more than starting every appliance when ready. The pipeline's mean latency was 18.5 seconds, and its API cost $0.0025 per recommendation.

**Table A.1 Robustness set: UK prices and tariff, December 2014 to July 2015 (48 scenarios; saving pooled over all runs).**

| System | Valid | Saving | Saving, strict limit | Saving, realistic limit |
|---|---|---|---|---|
| Multi-Agent Pipeline | 100 % | 6.69 % | 6.4 % | 6.9 % |
| Smart timer (own heuristic) | 100 % | 6.55 % | 6.2 % | 6.9 % |
| EMHASS, measured profiles | 100 % | 6.45 % | 6.0 % | 6.9 % |
| EMHASS, nominal power | 100 % | 6.06 % | 5.6 % | 6.5 % |
| Cheapest window | 70.8 % | 3.55 % | 1.5 % | 5.6 % |
| Cheapest-hours automation | 68.8 % | 2.62 % | 0.9 % | 4.3 % |
| LLM without tools | 95.1 % | 0.18 % | 0.3 % | 0.1 % |
| Optimum (upper bound) | 100 % | 6.69 % | 6.4 % | 6.9 % |

Paired per scenario, the pipeline saved more than the cheapest-hours automation (by 3.3 percentage points), the cheapest window (2.1) and the LLM without tools (6.4), with Wilcoxon $p < 0.001$ in each case, and it saved the same as EMHASS within $\pm 1$ percentage point (mean difference $+0.20$ points, 90 % confidence interval $[0.06, 0.34]$). These tests mirror the registered ones but were not registered for this set, so they are reported without adjustment.

The savings are about a quarter of those on the Finnish benchmarks. Because the price patterns are defined relative to the days of the window, even the volatile and spike days of 2015 are mild compared with the Finnish ones: on these days the spot price varied within the day by 8 to 9 cents per kWh on average, against 16 to 20 cents on the corresponding days of Q2. The window also has no negative prices. In addition, the constant charge makes up most of the consumer price, so a given shift in time changes a smaller share of the cost. The order of the price patterns also differs from the Finnish sets: the stable days gave the largest saving (9.8 % at the optimum) and the volatile days the smallest (3.0 %). The reason was not analysed further.

## A.3 Conclusion

Priced at the prices and tariff of their own country and period, the REFIT appliance cycles give much smaller savings, but the comparison between the systems does not change. The pipeline again reaches the optimal saving with a valid schedule every time, matches EMHASS, and saves more than the cheapest-hours automations, whose schedules again fail where the household's appliances must be staggered. The main results of this thesis therefore do not depend on combining UK appliance data with Finnish prices, although the size of the saving does depend on the price conditions.

# B. Evaluation of the earlier design

This appendix reports the evaluation of the earlier design of the study, summarised in Section 5.4. It compared the base multi-agent pipeline, without retrieval or the optimiser tool, with a single ReAct agent that had the first seven tools of Table 4.2 and the same step budget, on the first benchmark of 48 scenarios in six types (Section 5.1) and on the held-out Q set (Section 5.2). Its outcome was the optimality gap, the cost of a schedule above the optimal cost in percent of the optimal cost, with an infeasible schedule scored at the cost of starting every appliance when ready. The research questions and hypotheses of this design were replaced in the redesign, so the labels SQ1, SQ2, H1 and H2 in Section B.3 refer to the first pre-registration. The results remain valid for that design.

## B.1 Pipeline and single agent on the first benchmark

This section reports the tests of the earlier design for GPT-4o-mini on the first benchmark, with the mean over three runs per scenario as the paired observation ($N = 48$). These results are exploratory; the pre-registered confirmatory results are reported in Section B.3. A run with the local model gpt-oss-20b was stopped before completion and is not reported.

**Schedule quality.** The multi-agent pipeline reached a mean optimality gap of 1.25 % and the single agent 1.43 %. The Wilcoxon signed-rank test finds no significant difference ($W = 449$, $Z = -0.14$, $p = 0.89$, $r = 0.02$), and the median paired difference is zero, because both designs reach the same cost in most scenarios. The equivalence test shows that this is not merely a lack of evidence. The mean paired difference is $-0.18$ percentage points, with a 90 % confidence interval of $[-0.71, 0.35]$, which lies inside the primary margin of $\pm 1$ percentage point (Table B.1). The two designs are therefore equivalent in schedule quality within one percentage point of the optimum. At the narrower margin of 0.5 percentage points the $t$ version of the test does not conclude equivalence, because a few scenarios with large differences in either direction widen the interval, although the Wilcoxon version does.

**Table B.1 Equivalence test (TOST) of the optimality gap, multi-agent vs. single-agent (GPT-4o-mini, $N = 48$).**

| Margin | $p$, paired $t$ TOST | $p$, Wilcoxon TOST | Equivalent |
|---|---|---|---|
| $\pm 0.5$ percentage points | 0.156 | $< 0.001$ | No |
| $\pm 1$ percentage point (primary) | 0.006 | $< 0.001$ | Yes |
| $\pm 2$ percentage points | $< 0.001$ | $< 0.001$ | Yes |

**Efficiency.** The two designs differ significantly in every efficiency measure (Table B.2). The pipeline used 34 % fewer input tokens, because each specialist works with a short context and only its own tools, while the single agent carries the entire conversation and all eight tool descriptions through every step. It produced 83 % more output tokens and made 23 % more tool calls, since four agents each write their own structured output and call their tools separately. Input tokens dominate the price, so the API cost per run was 21 % lower for the pipeline, and lower in 35 of the 48 scenarios. The sequential hand-offs made the pipeline 69 % slower, with a mean of 21.4 seconds against 12.7 seconds, and it was faster in only two scenarios.

**Table B.2 Efficiency per run, multi-agent vs. single-agent (GPT-4o-mini; tokens and tool calls per run, API cost in USD, latency in seconds; each value is the mean over $N = 48$ scenarios of the per-scenario mean of three runs; Wilcoxon signed-rank test).**

| Measure | Single agent | Multi-agent | Difference | Multi-agent lower | $p$ | $r$ |
|---|---|---|---|---|---|---|
| Input tokens | 21 686 | 14 292 | $-34\,\%$ | 43/48 | $< 0.001$ | 0.79 |
| Output tokens | 661 | 1 208 | $+83\,\%$ | 1/48 | $< 0.001$ | 0.87 |
| Total tokens | 22 347 | 15 500 | $-31\,\%$ | 40/48 | $< 0.001$ | 0.75 |
| API cost (USD) | 0.0037 | 0.0029 | $-21\,\%$ | 35/48 | $< 0.001$ | 0.54 |
| Latency (s) | 12.7 | 21.4 | $+69\,\%$ | 2/48 | $< 0.001$ | 0.86 |
| Tool calls | 16.5 | 20.3 | $+23\,\%$ | 1/48 | $< 0.001$ | 0.81 |

Taken together, decomposing the task into specialist agents did not lower the cost of the schedules, but it reached the same quality with about a third fewer tokens and a fifth lower inference cost, at the price of a longer response time.

## B.2 Comparison with baselines and real tools

Table B.3 compares the two agent designs with the smart timer, the LLM without tools and the two real home energy management tools. All systems that use a language model use GPT-4o-mini.

Without tools, GPT-4o-mini schedules poorly. Its mean optimality gap is 23.1 %, and 18 of its 144 schedules are infeasible: 14 exceed the peak limit and 4 start a cycle outside its window. A further 22 of its feasible schedules cost more than starting every appliance as soon as it is ready, and over all its runs it captures only 13 % of the achievable saving, against about 90 % for the agents. Both agent designs are significantly better (Wilcoxon $p < 0.001$, $r = 0.87$ for each), with median paired differences of 10.8 percentage points for the single agent and 11.4 for the pipeline. The same model that fails to schedule from a single prompt comes close to the optimum when it can query prices, costs and loads through tools and check its own work. Tool use, rather than the language model's own calculation, is therefore what makes the agents viable.

**Table B.3 Agent designs, simpler baselines and real home energy management tools (GPT-4o-mini, main benchmark, 48 scenarios).**

| System | Feasible runs | Mean gap | Median gap | Gap, type D | Savings captured | Time per run |
|---|---|---|---|---|---|---|
| Smart timer (own heuristic) | 48/48 | 0.23 % | 0.00 % | 1.2 % | 99 % | $<$ 0.01 s |
| EMHASS, profiles | 48/48 | 0.32 % | 0.00 % | 0.1 % | 99 % | 1.2 s |
| Multi-agent pipeline | 144/144 | 1.25 % | 0.01 % | 5.9 % | 93 % | 21.4 s |
| Single agent | 143/144 | 1.43 % | 0.00 % | 6.5 % | 90 % | 12.7 s |
| EMHASS, nominal power | 48/48 | 2.24 % | 0.45 % | 8.5 % | 87 % | 1.0 s |
| Cheapest hours | 25/48 | 17.6 % | 4.2 % | 62.7 % | 25 % | $<$ 0.01 s |
| LLM without tools | 126/144 | 23.1 % | 11.9 % | 66.0 % | 13 % | 2.0 s |

The smart timer, however, outperforms both agent designs. It finds a feasible schedule in every scenario, is optimal in 30 of the 48, and its largest gap is 3.7 % (D04). Both agent designs have significantly larger gaps than the timer (single agent $p < 0.001$, $r = 0.63$; pipeline $p < 0.001$, $r = 0.68$). The difference is concentrated in type D, where the timer's mean gap is 1.2 % against 5.9 % for the pipeline and 6.5 % for the single agent. The median paired differences are small (0.002 percentage points for the single agent and 0.017 for the pipeline), because in most scenarios all three systems reach the optimum or come very close to it. The timer needs no language model, runs in about two milliseconds and has no inference cost. In this benchmark, where the power profile and window of every appliance are known exactly and the horizon is one day, placing the appliances one at a time is almost always enough. The agents come within about one percentage point of this rule but do not improve on it. What they add is the ability to start from a request written in natural language rather than from structured appliance data, and to explain the schedule they produce.

The two real home energy management tools, added after the pre-registration, bracket the agents. Configured with the measured power profiles, EMHASS comes close to the optimum: every schedule is feasible, 32 of the 48 are optimal, and the mean gap is 0.32 %. Its largest gap, 4.2 % in F02, comes from its default solver tolerance of 1 % (the setting `lp_solver_mip_rel_gap` = 0.01; EMHASS contributors, 2026), which applies to the whole electricity bill including the base load and can therefore amount to several percent of the appliance cost scored here. With an exact tolerance the same optimization comes within 0.2 % of the optimum in every scenario, which confirms that the MILP reference reflects what a deployed open-source optimizer can achieve. The agents' mean gaps are about one percentage point higher, but the difference from EMHASS as deployed is not significant (Wilcoxon $p = 0.24$ for the single agent and $p = 0.19$ for the pipeline). Configured with a constant nominal power, EMHASS still keeps every schedule within the peak limit, because it plans each cycle at its peak power, but it misjudges the cost of the real cycles and reaches a mean gap of 2.2 %. Both agent designs are significantly better than this configuration (Wilcoxon $p < 0.001$ for each, $r = 0.67$ for the single agent and $0.64$ for the pipeline). The cheapest-hours automation, under the assumption stated in Section 3.3 that its trigger starts a cycle which then runs to completion, exceeds the peak

limit in 23 of the 48 scenarios, because each appliance follows its own cheap period and appliances whose windows share a price minimum start together. Scored at the naive cost when infeasible, it reaches a mean gap of 17.6 %, and both agent designs are significantly better ($p < 0.001$, $r = 0.82$ and $0.81$). The agents are therefore outperformed by the smart timer and are not significantly different from a well-configured optimizer as deployed, while they outperform a simply configured optimizer and the per-device automation that is common among households.

Two checks show that this ranking does not depend on how the costs are scored. First, scoring an infeasible run at the cost of the naive schedule does not drive the result. Restricted to feasible runs, the mean gaps are 0.23 % for the smart timer, 0.32 % for EMHASS with profiles, 1.25 % for the pipeline, 1.45 % for the single agent, 2.24 % for EMHASS with nominal power, 7.9 % for the cheapest-hours automation and 23.0 % for the LLM without tools, the same order as in Table B.3, and the order on the held-out benchmark is also unchanged. The cheapest-hours automation looks better on feasible runs only because these are the days on which its appliances happen not to collide. Second, the ranking is the same under the spot price alone as under the consumer price, as explained in Section 3.3.

## B.3 Confirmatory run on the held-out Q set

The hypotheses of the earlier SQ1 and SQ2 were fixed in the pre-registration, which was committed to the project repository on 30 September 2026 together with the code, prompts, tools and analysis script. The confirmatory run followed on the held-out benchmark of 48 scenarios with fifteen-minute prices (Section 5.2), on which no language model had been run before. All systems that use a language model used GPT-4o-mini with three runs per scenario, and the analysis was run exactly as registered. The run cost USD 0.92 in API fees. Table B.4 reports the six registered tests.

**Table B.4 Pre-registered confirmatory tests on the held-out fifteen-minute benchmark (GPT-4o-mini, $N = 48$; $p$-values Holm-adjusted within each sub-question).**

| | Hypothesis | Estimate | $p$ | $r$ | Supported |
|---|---|---|---|---|---|
| H1 | Single agent gap < LLM without tools | $-17.9$ pp | $< 0.001$ | 0.86 | Yes |
| H1 | Pipeline gap < LLM without tools | $-17.7$ pp | $< 0.001$ | 0.86 | Yes |
| H2a | Pipeline equivalent to single agent ($\pm 1$ pp) | $+0.22$ pp | $< 0.001$ | – | Yes |
| H2b | Pipeline uses fewer input tokens | $-31$ % | $< 0.001$ | 0.84 | Yes |
| H2b | Pipeline has a lower API cost per run | $-21$ % | $< 0.001$ | 0.70 | Yes |
| H2c | Pipeline has a longer latency | $+52$ % | $< 0.001$ | 0.80 | Yes |

All six hypotheses are supported. For H1, both agent designs reached a mean optimality gap of about one percent (0.86 % for the single agent and 1.08 % for the pipeline), whereas the LLM without tools reached 18.8 %. Fifteen of its 144 schedules were infeasible, and over all its runs it captured only 24 % of the achievable saving, against 92 % for the single

agent and 90 % for the pipeline. For H2a, the mean paired difference in gap was $+0.22$ percentage points with a 90 % confidence interval of $[-0.15, 0.59]$, which lies well inside the margin of $\pm 1$ percentage point. On this benchmark the pipeline was thus marginally worse than the single agent on average, the reverse of the exploratory result, but in both runs the difference is small enough to count as equivalent. For H2b and H2c, the pipeline used 13 310 input tokens per run against 19 189 for the single agent, its API cost was 21 % lower, and its latency was 20.4 seconds against 13.4 seconds.

The confirmatory estimates are close to the exploratory ones: input tokens fell by 31 % in both runs, API cost by 21 % in both, and latency rose by 52 % here against 69 % before. The secondary results also replicate. The equivalence test is again not significant at the narrower margin of $\pm 0.5$ percentage points in its $t$ version ($p = 0.11$), although its Wilcoxon version is. The smart timer again outperforms both agent designs, with a mean gap of 0.17 % (Wilcoxon $p < 0.001$ against each design, $r = 0.69$ for the single agent and $0.64$ for the pipeline). Type D remains the hardest scenario type, with mean gaps of 3.9 % for the single agent and 5.5 % for the pipeline, while the LLM without tools fails most clearly in type E, with a mean gap of 56 %.

The real home energy management tools were added after the pre-registration, so the following comparisons are secondary and not registered; their $p$-values are descriptive and not adjusted for multiple testing. EMHASS as deployed reaches a mean gap of 0.36 % with measured profiles, not significantly different from either agent design (Wilcoxon $p = 0.90$ for the single agent and $p = 0.69$ for the pipeline), and 3.5 % with nominal power, significantly worse than both ($p < 0.001$). The cheapest-hours automation exceeds the peak limit in 26 of the 48 scenarios, with a mean gap of 18.7 %. These patterns match those on the main benchmark.

The registered findings, H1 for SQ1 and H2 for SQ2, are therefore not specific to the benchmark on which the pipeline was developed: they hold on unseen days with a different price resolution. They cover GPT-4o-mini only. The registered hypothesis on a small model run locally (H3) was withdrawn in the first amendment of the pre-registration, before its run on the first benchmark was completed.